Working Report

Heat and Cold Deaths Under Four Climate Scenarios, 2022–2050: A Multi-Model Summary



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Abstract

We ask how many deaths and years of life lost may be attributable to non-optimal temperatures between 2022 and 2050 under four climate scenarios, and how much of what can be said about those numbers survives disagreement between climate models and uncertainty in the underlying risk estimates. We applied the Global Burden of Disease exposure-response curves to daily temperature from 24 climate models in 204 countries, scaled each scenario to IHME's mortality forecasts, and pooled every combination of climate model and uncertainty draw (12,000 values per number) into one interval. Under SSP2-4.5, world heat deaths rise from 431,244 per year in 2022–2030 (95% interval 328,460 to 570,407) to 729,507 in 2041–2050 (499,064 to 1,074,365), and cold deaths from 1,483,673 (1,302,682 to 1,661,329) to 1,998,935 (1,602,802 to 2,423,909). Most of this rise occurs in every scenario: by 2041–2050, SSP5-8.5 has 1.127 times the heat deaths of SSP2-4.5 and 0.976 times the cold deaths, and the two nearly offset, so that total temperature-attributable deaths differ by a ratio of 1.016 with an interval on the difference that includes zero. These numbers rest on IHME's forecasts of population and mortality and on curves that do not change over time, and they are not forecasts of what will happen; they do not test adaptation, and the intervals do not include every source of uncertainty.

1 Introduction

The World Bank's Global Climate-Health Program wishes to know what future cause-specific mortality attributable to temperature could look like under different climate scenarios, holding non-temperature risks at their forecast values. Our aim here is to summarize the finished scenario runs in the form agreed at the team meeting of October 27, 2026: one number and one set of uncertainty intervals for each quantity, built on a common set of climate models, showing heat, cold and their sum side by side, with breakdowns by cause, age and sex, and income group, as averages over decades rather than single years.

The rest of this report is organized as follows. Section 2 describes how the numbers are constructed. Section 3 gives the results: world totals, the differences between scenarios, where the uncertainty comes from, income groups, countries (including two that we mark with an asterisk), causes, age and sex, years of life lost, the year heat overtakes cold, and the sensitivity to the choice of climate models. Section 4 says how the numbers were checked, and Section 5 sets out what we would not claim from them. Country tables are in the appendices.

2 Data and methods

2.1 Attributable deaths

We follow the approach of Burkart et al. (Lancet, 2021) and the Global Burden of Disease (GBD) study. For each location, 17 causes of death are linked to temperature through cause-specific exposure-response curves, which give the relative risk of death at a given daily temperature within a temperature zone defined by a pixel's annual mean temperature. A pixel's zone is reassigned each year from that year's mean, but the curves themselves do not change over time. Each cause-specific population attributable fraction (PAF), the share of that cause's deaths attributed to temperature, is computed against a theoretical minimum risk exposure level (TMREL), the temperature of lowest death-weighted risk, which we derive separately for each uncertainty draw and year from that draw's curves and cause-of-death mix. Pixel-level results are weighted by population to the national level. We use 500 uncertainty draws, and the same draw index d refers to the same draw of the curves and of the mortality forecast across years, countries, scenarios and climate models, which is what allows the differences below to be taken draw by draw.

Daily temperature comes from the World Bank Climate Change Knowledge Portal's downscaled and bias-corrected CMIP6 product at 0.25 degrees. We use four Shared Socioeconomic Pathway (SSP) scenarios, SSP1-2.6, SSP2-4.5, SSP3-7.0 and SSP5-8.5, for 2022–2050. Deaths come from IHME's GBD 2021 Foresight reference forecast, which IHME defines under SSP2-4.5.

2.2 Scaling to IHME's forecasts

Because IHME's mortality forecast already embeds SSP2-4.5 temperature, deaths under another scenario are obtained by removing that temperature component and applying the scenario's own (the step we have called Workflow B, which is the approach set out in the World Bank's data request to IHME). For cause c and draw d, with m the IHME forecast deaths and PAF the non-optimal attributable fraction,

$$m^{\text{scaled}}_{c,d} = m_{c,d} \times \frac{S^{\text{target}}_{c,d}}{S^{\text{SSP2-4.5}}_{c,d}}, \qquad S = \frac{1}{1 - \text{PAF}},$$

for the 13 causes in which IHME's forecast carries a temperature scalar; for the remaining four (cardiomyopathy and myocarditis, animal contact, forces of nature, and other unintentional injuries) the ratio is set to 1. Attributable deaths are then m^{scaled} times the scenario's PAF. SSP2-4.5 scaled against itself reproduces the unscaled burden, and the team chose to keep this step for the headline numbers; at world level it moves totals by less than 0.15%, so it matters little for what follows.

2.3 Pooling across climate models

Each number in this report is computed from 12,000 values, the 24 climate models run under all four scenarios crossed with 500 draws. An interval is taken from the quantiles of that pool, so disagreement between climate models is part of it, and each model carries equal weight. Reported intervals are 95% (the GBD convention), 80%, and 66% (the IPCC's "likely" range). Sums over locations or years are taken within a draw and model before pooling, and a difference between scenarios is taken draw by draw within a model before pooling, so that the uncertainty shared by the two scenarios does not enter the difference. Decades are 2022–2030 (nine years), 2031–2040 and 2041–2050, and every value is per year.

The 24 models are not 24 independent results: several share code and history (two EC-Earth,

two MPI, two NorESM, two INM, two CNRM, two MIROC, and the HadGEM and UKESM family). Pooled intervals treat the models as a sample, and we would not read them as probabilities in any strict sense.

2.4 Measures

Rates are crude deaths per 100,000 people per year, with population taken from the mean of IHME's forecast draws for the slice in question, averaged over the decade's years; because population is held at that mean, a rate's interval reflects the deaths only. Crude rates are not age-standardized, so a rise can reflect an older population and not only a higher risk at each age. Years of life lost (YLLs) use our national definition, deaths times the country's UN World Population Prospects remaining life expectancy at the start of the 5-year age group (80+ uses age 80). These are not comparable with GBD's YLLs, which use a reference life table. Income groups are the World Bank's, from its country list as downloaded on November 6, 2026, applied to all years: 71 high income, 58 upper middle, 47 lower middle and 25 low income. Taiwan is absent from that list and is taken as high income from the Bank's classification spreadsheet, and Cook Islands, Niue and Tokelau appear in neither and stay unclassified (they sum to about one death per year). This mapping has not yet been reviewed by a second person.

3 Results

3.1 How many deaths, and how do they change over time?

Table 1 gives world totals for each scenario and decade, and Figure 1 shows the same numbers with the three interval levels. Under SSP2-4.5, heat deaths rise by 69% between 2022–2030 and 2041–2050 and cold deaths by 35%, and heat's share of the two together rises from 22.5% to 26.7%, so cold remains the larger burden throughout the period in every scenario. As a rate, heat deaths go from 5.25 to 7.89 per 100,000 people per year and cold from 18.06 to 21.63, smaller increases than in counts because population grows from 8.22 to 9.24 billion.

Table 1. World deaths per year attributable to heat and cold, 24 common climate models, by scenario and decade: pooled mean and 95% interval.

Period	Scenario	Heat deaths	Cold deaths	Heat + cold
2022–2030	SSP1-2.6	439,564 (335,071 to 577,145)	1,476,698 (1,300,283 to 1,649,518)	1,916,262 (1,710,301 to 2,108,140)
	SSP2-4.5	431,244 (328,460 to 570,407)	1,483,673 (1,302,682 to 1,661,329)	1,914,917 (1,709,009 to 2,106,891)
	SSP3-7.0	424,074 (317,094 to 563,785)	1,493,569 (1,309,047 to 1,672,803)	1,917,644 (1,708,095 to 2,107,858)
	SSP5-8.5	438,032 (337,926 to 566,727)	1,479,161 (1,302,715 to 1,651,236)	1,917,193 (1,712,835 to 2,107,675)
2031–2040	SSP1-2.6	559,818 (406,659 to 783,095)	1,732,357 (1,464,364 to 2,010,174)	2,292,174 (1,953,869 to 2,641,967)
	SSP2-4.5	550,839 (401,464 to 763,498)	1,740,378 (1,473,028 to 2,020,121)	2,291,217 (1,955,573 to 2,640,692)
	SSP3-7.0	555,087 (401,912 to 810,299)	1,750,587 (1,475,667 to 2,036,943)	2,305,673 (1,964,279 to 2,664,307)
	SSP5-8.5	586,214 (411,226 to 827,779)	1,724,286 (1,455,503 to 2,010,306)	2,310,500 (1,967,530 to 2,671,937)
2041–2050	SSP1-2.6	703,149 (482,305 to 1,014,120)	2,005,660 (1,610,224 to 2,429,040)	2,708,809 (2,185,166 to 3,285,540)
	SSP2-4.5	729,507 (499,064 to 1,074,365)	1,998,935 (1,602,802 to 2,423,909)	2,728,442 (2,197,956 to 3,315,938)
	SSP3-7.0	735,317 (495,059 to 1,116,333)	2,006,773 (1,607,553 to 2,435,824)	2,742,090 (2,209,346 to 3,336,578)
	SSP5-8.5	821,948 (542,817 to 1,243,627)	1,950,759 (1,559,517 to 2,374,196)	2,772,707 (2,224,938 to 3,394,792)

Why do cold deaths rise when the world is warming? The rise follows IHME's forecast of deaths from these 17 causes, which grows from 32.0 to 43.4 million per year between the two decades. Cold's share of those deaths stays nearly flat (4.66% to 4.62%), while heat's rises (1.32% to 1.69%). This is

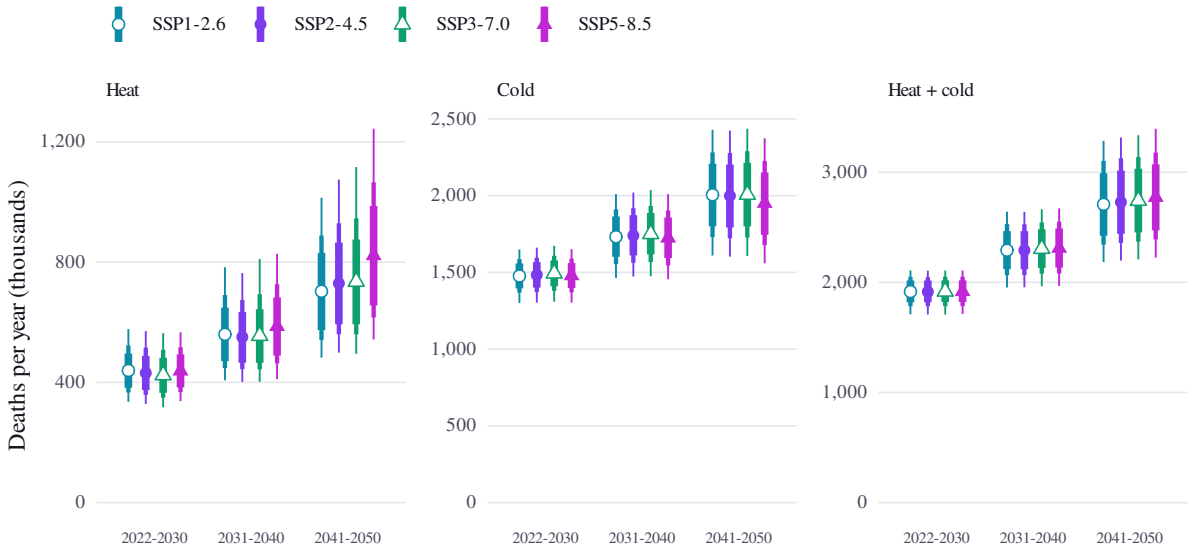


Figure 1. World deaths per year by decade and scenario, 24 common climate models. The point is the pooled mean; the thin, medium and thick bars are the 95%, 80% and 66% intervals. The vertical scales differ between panels.

consistent with an aging population moving into the causes and ages most affected by cold, at about the rate that warming reduces cold exposure, but we have not split this further. An earlier split of the rise in heat deaths, on single years under SSP2-4.5 (2022 to 2050, as model means, without uncertainty; `docs/reviews/heat-rise-decomposition-2026-10-27.org`), attributes the +106% to population size (+23 percentage points), aging (+106), lower death rates (−70) and a higher heat share (+47), which suggests that most of the rise in heat deaths would occur without further warming, as far as that split reaches.

Figure 2 shows the same series by year. It is shown to make clear why we do not report single years: world heat deaths within one model change by −19% to +29% from one year to the next (1st to 99th percentile over runs), and by as much as +47% in one case (MIROC-ES2L under SSP3-7.0 in 2032, of which India accounts for 77,100 of the 176,124 added deaths). Such spikes are not shared across models, which is consistent with simulated weather and not with an error in a common input.

3.2 How much do the scenarios differ?

Table 2 and Figure 3 set each scenario against SSP2-4.5. The clearest departure is SSP5-8.5 in 2041–2050, with 92,442 more heat deaths per year (95% interval 24,344 to 206,886; ratio 1.127, and higher than SSP2-4.5 in all 24 models) and 48,176 fewer cold deaths (−106,725 to −16,804; ratio 0.976, lower in all 24). Because these offset, the total differs by 44,266 (−14,239 to 141,455; ratio 1.016), and 21 of 24 models have a higher total. For SSP1-2.6 in the same decade, heat is lower by 26,357 (−106,815 to 36,835; ratio 0.964, with 7 of 24 models above SSP2-4.5), an interval which includes zero. SSP3-7.0 cannot be told apart from SSP2-4.5 by this measure (heat ratio 1.008, 13 of 24 models above).

Before 2040 the order is not the one a reader might expect: SSP1-2.6 has slightly more heat deaths than SSP2-4.5 (ratios 1.019 and 1.016, with 17 and 19 of 24 models above) and SSP3-7.0 slightly fewer in 2022–2030 (0.983, 7 of 24). Temperature is consistent with this: in a death-weighted average over countries, SSP1-2.6 is 0.11°C warmer than SSP2-4.5 in 2030 and 0.25°C cooler by 2050 (from the four

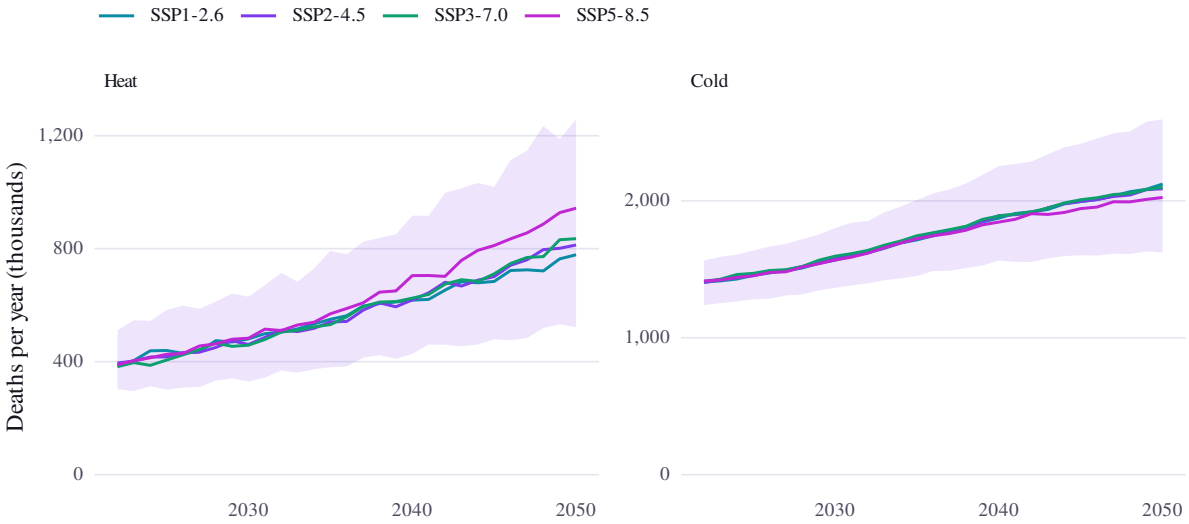


Figure 2. World deaths per year by single year: mean over the 24 common climate models and 500 draws, for each scenario. The shaded band is the pooled 95% interval for SSP2-4.5. Single years are shown only to illustrate variability; we report decades.

Table 2. Target scenario against SSP2-4.5, world deaths per year, 24 common models. Ratio is target mean over SSP2-4.5 mean. Difference is taken draw by draw within a model and then pooled: mean and 95% interval.

Scenario	Period	Heat		Cold		Heat + cold	
		Ratio	Difference	Ratio	Difference	Ratio	Difference
SSP1-2.6	2022–2030	1.019	8,319 (−42,673 to 46,197)	0.995	−6,975 (−39,159 to 27,441)	1.001	1,345 (−35,349 to 31,469)
	2031–2040	1.016	8,979 (−27,089 to 41,594)	0.995	−8,021 (−32,357 to 22,077)	1.000	958 (−30,482 to 25,969)
	2041–2050	0.964	−26,357 (−106,815 to 36,835)	1.003	6,725 (−61,798 to 57,261)	0.993	−19,633 (−76,956 to 27,716)
SSP3-7.0	2022–2030	0.983	−7,170 (−27,939 to 17,265)	1.007	9,896 (−11,573 to 36,351)	1.001	2,726 (−20,648 to 29,317)
	2031–2040	1.008	4,248 (−43,003 to 63,395)	1.006	10,209 (−24,716 to 45,838)	1.006	14,457 (−20,924 to 52,071)
	2041–2050	1.008	5,810 (−82,064 to 86,069)	1.004	7,838 (−53,872 to 50,423)	1.005	13,648 (−42,317 to 64,657)
SSP5-8.5	2022–2030	1.016	6,788 (−33,251 to 49,549)	0.997	−4,512 (−39,585 to 29,931)	1.001	2,276 (−40,236 to 33,478)
	2031–2040	1.064	35,375 (−21,707 to 88,748)	0.991	−16,092 (−49,508 to 25,754)	1.008	19,284 (−17,112 to 60,293)
	2041–2050	1.127	92,442 (24,344 to 206,886)	0.976	−48,176 (−106,725 to −16,804)	1.016	44,266 (−14,239 to 141,455)

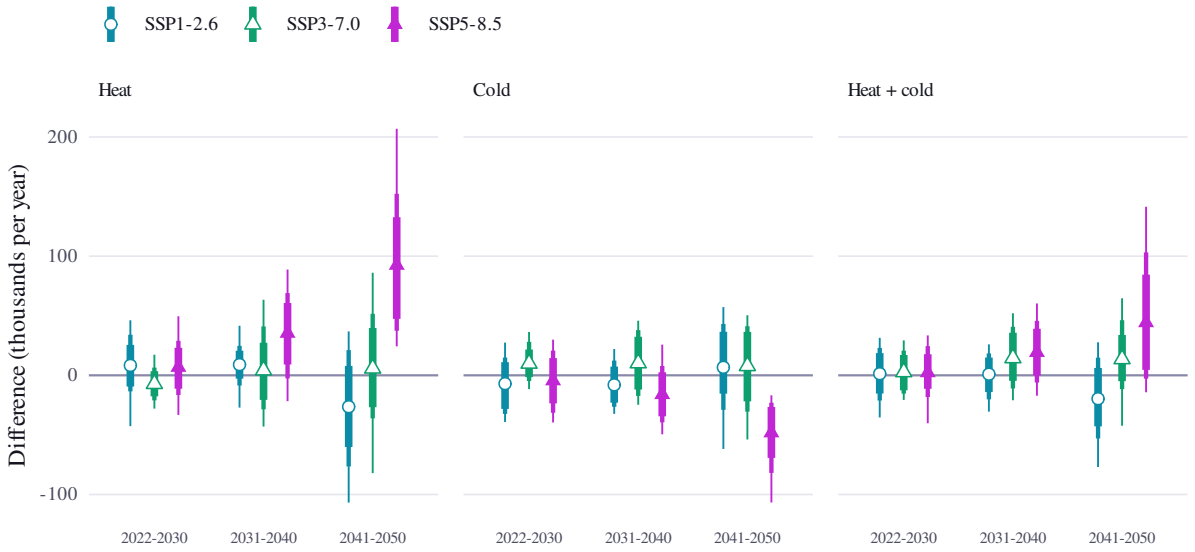


Figure 3. Difference from SSP2-4.5 in world deaths per year, by decade, with the 95%, 80% and 66% intervals (thin, medium, thick bars) of the draw-by-draw difference, 24 common climate models.

years we hold temperature for; review of October 26, 2026). One possibility is the faster reduction of cooling aerosols under SSP1-2.6, which we have not tested.

At country level the same picture holds with less power. The 95% interval for the SSP5-8.5 heat difference in 2041–2050 is entirely above zero in 39 of 204 countries and entirely below in none. For SSP1-2.6 it is entirely below zero in one country and for SSP3-7.0 in none. With 204 comparisons, some intervals would exclude zero even with no underlying difference, so these counts are not rates, and the differences between countries should not be overread.

3.3 Where does the uncertainty come from?

For world heat deaths, about half of the pooled variance is between climate models (0.41 to 0.57, by scenario and decade), and the remainder comes from the draws of the curves and the mortality forecast. For cold, the share between models is about a tenth (0.07 to 0.14). For heat plus cold it is only 0.02 to 0.04, because a model that runs warm has more heat deaths and fewer cold deaths. The sum is thus much less sensitive to the choice of climate model than either part, and its interval (coefficient of variation about 0.11 in 2041–2050) is carried almost entirely by the exposure-response curves and the mortality forecast. Table 3 gives the three interval levels and the range of the 24 model means for the most-compared cells.

Table 3. World deaths per year, 2041–2050, 24 common models: pooled mean, 95%, 80% and 66% intervals, and the lowest and highest of the 24 model means.

Scenario	Measure	Mean	95%	80%	66%	Range of model means
SSP2-4.5	Heat	729,507	499,064 to 1,074,365	560,231 to 928,985	593,357 to 865,107	613,078 to 950,392
SSP2-4.5	Cold	1,998,935	1,602,802 to 2,423,909	1,722,527 to 2,278,352	1,792,883 to 2,202,615	1,841,369 to 2,091,710
SSP2-4.5	Heat + cold	2,728,442	2,197,956 to 3,315,938	2,357,929 to 3,127,341	2,440,927 to 3,015,052	2,640,495 to 2,834,469
SSP5-8.5	Heat	821,948	542,817 to 1,243,627	615,786 to 1,065,728	655,970 to 986,615	645,102 to 1,076,591
SSP5-8.5	Cold	1,950,759	1,559,517 to 2,374,196	1,677,583 to 2,226,376	1,744,700 to 2,151,321	1,801,681 to 2,058,742
SSP5-8.5	Heat + cold	2,772,707	2,224,938 to 3,394,792	2,389,996 to 3,178,995	2,472,950 to 3,071,703	2,672,316 to 2,891,928

3.4 How do income groups differ?

Table 4 and Figure 4 give the income groups. The burden differs in kind. Cold rates under SSP2-4.5 in 2041–2050 are high in the high-income (40.2 per 100,000 per year) and upper-middle-income (35.1) groups and low in the lower-middle (7.5) and low-income (5.4) groups, while heat rates are highest in the lower-middle-income group (10.7), compared with 7.4, 5.2 and 4.1 in the upper-middle, low and high-income groups. These are crude rates, and much of the contrast in cold is likely age structure: high-income countries are older, and most cold deaths occur at 65 and above. Low- and middle-income countries hold 92% of world heat deaths in that decade and 71% of cold deaths. The proportional effect of SSP5-8.5 on heat is similar across the wider groups (1.109 in high-income and 1.128 in low- and middle-income countries), but the difference in deaths is not: 6,446 and 85,995 per year.

Table 4. Deaths per year and crude rate per 100,000 per year (95% interval), by World Bank income group, SSP2-4.5, 2041–2050, 24 common models, and the SSP5-8.5 to SSP2-4.5 ratio of means. World equals the four income groups plus three unclassified locations.

Group (countries)	Heat, SSP2-4.5		Cold, SSP2-4.5		SSP5-8.5 / SSP2-4.5	
	Deaths	Rate (95%)	Deaths	Rate (95%)	Heat	Cold
World (1)	729,507	7.9 (5.4 to 11.6)	1,998,935	21.6 (17.3 to 26.2)	1.127	0.976
High income (71)	59,262	4.1 (2.8 to 6.1)	578,046	40.2 (32.0 to 46.6)	1.109	0.995
Low and middle income (1)	670,244	8.6 (5.8 to 12.7)	1,420,889	18.2 (14.3 to 22.7)	1.128	0.968
Upper middle income (58)	231,191	7.4 (4.9 to 11.8)	1,094,165	35.1 (27.9 to 43.1)	1.168	0.973
Lower middle income (47)	380,137	10.7 (7.0 to 16.2)	264,871	7.5 (5.0 to 10.3)	1.111	0.945
Low income (25)	58,916	5.2 (3.2 to 7.8)	61,852	5.4 (4.2 to 6.9)	1.086	0.973



Figure 4. Crude deaths per 100,000 people per year by World Bank income group, 24 common climate models: SSP2-4.5 in 2022–2030 and 2041–2050, and SSP5-8.5 in 2041–2050. The point is the pooled mean; thin, medium and thick bars are the 95%, 80% and 66% intervals. The vertical scales differ between panels.

3.5 Which countries, and which need a caution?

The world totals are concentrated. In 2041–2050 under SSP2-4.5, India accounts for 31.7% of heat deaths (231,555 per year, 95% interval 146,386 to 359,901) and China for 39.6% of cold deaths (792,478, from 621,344 to 978,559), and the five largest countries hold 58.6% of heat deaths and 58.0% of cold deaths. Figure 5 maps the rates, and Table 11 in Appendix A lists every country.

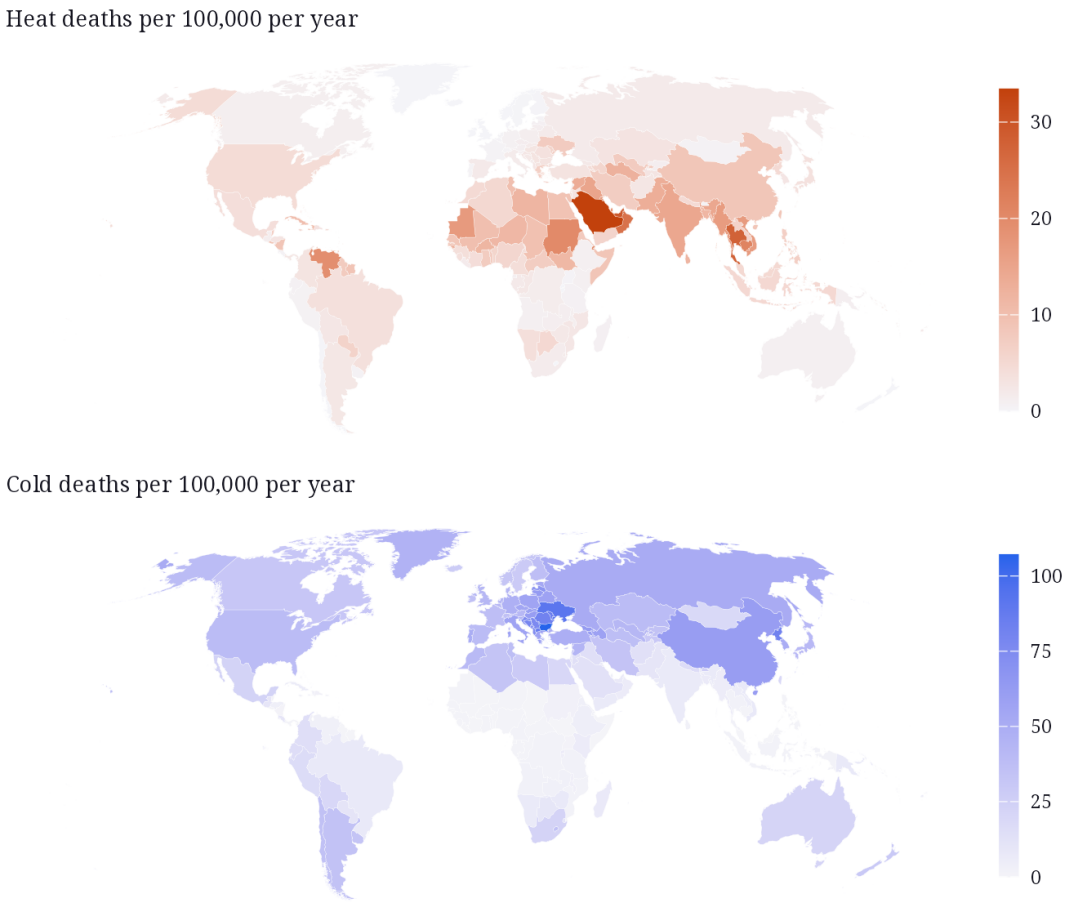


Figure 5. Crude heat and cold deaths per 100,000 people per year, by country, SSP2-4.5, 2041–2050, pooled mean over the 24 common climate models. Each panel has its own scale. The maps do not show uncertainty, and small island states are barely visible; see Table 11.

Two cautions apply to reading the country numbers. The first concerns the Gulf states, which lead the heat rate table: Saudi Arabia 33.4 per 100,000 per year (18.6 to 54.1), United Arab Emirates 29.2 (14.5 to 47.6). IHME's forecast multiplies 17-cause deaths in the United Arab Emirates by 10.6 between 2022 and 2050 (10,961 to 116,058), as well as by 8.2 in Qatar, 4.3 in Oman, 4.1 in Bahrain and Kuwait, and 3.2 in Saudi Arabia, against 1.52 for the world, and its population aged 65 and over in the

United Arab Emirates goes from 198,137 to 4,717,387. This is consistent with migrant workers being forecast to age in place, which is our guess and not something the files state. The pipeline follows these inputs as given, so the Gulf rates in 2041–2050 depend on that forecast.

The second concerns disasters. IHME's forecast draws for the cause "exposure to forces of nature" include, in a few draws, deaths on the scale of a major event. A PAF of a percent or two on those deaths swings a small country's total, and in Haiti such draws also pull the derived TMREL to the bottom of its search range, which then applies to all 17 causes in that draw (review of September 14, 2026). We keep all 17 causes in every headline number, as the team decided, and mark the two countries where this matters most, Haiti and Honduras, with an asterisk, showing the totals beside the same totals without that cause (Table 5). The column without the cause removes only that cause's own deaths and not its effect on the TMRELs, so it does not undo the second mechanism in Haiti, and it should not be read as a corrected estimate: Haiti's interval without the cause includes zero, which we suspect is for that reason. For Haiti, the mean is a poor summary of the distribution; the median non-optimal value in a single year is 539 deaths where the highest is 115,656. Across all countries under SSP2-4.5 in 2041–2050, the upper bound of the 95% interval moves by more than 2% in 8 of 204 and by more than 10% only in these two. The effect on larger aggregates is small: in 2041–2050, 645,009 against 642,870 deaths per year for the lower-middle-income group, in which both countries fall, and 2,728,442 against 2,727,440 for the world. Single draws show the same feature elsewhere (19,041 deaths in one year of one draw in Puerto Rico, against a median of 129; 81,968 in Myanmar, against 11,516), which is a further reason to avoid single-year country values, although in these cases decade means and intervals are little affected.

Table 5. The two countries marked with an asterisk: heat plus cold deaths per year under SSP2-4.5, 24 common models, mean and 95% interval, with all 17 causes (the headline number) and without the cause "exposure to forces of nature".

Country	Period	All 17 causes (95%)	Without forces of nature (95%)	Ratio of means
Haiti*	2022–2030	2,084 (282 to 10,715)	236 (−713 to 603)	8.82
	2041–2050	2,032 (417 to 8,575)	497 (−461 to 1,093)	4.09
Honduras*	2022–2030	547 (360 to 954)	407 (145 to 594)	1.34
	2041–2050	818 (530 to 1,216)	702 (349 to 1,051)	1.17

3.6 Which causes?

Table 6 and Figure 6 give the causes for 2041–2050 under SSP2-4.5. Ischemic heart disease and stroke are the largest for heat (168,892 and 144,203 deaths per year, together 43% of the total), followed by diabetes, chronic obstructive pulmonary disease (COPD) and lower respiratory infections. For cold, ischemic heart disease leads (562,182), followed by COPD (475,014) and stroke (413,001). Diabetes is unusual in having about equal heat and cold deaths (85,485 and 84,124). The nine injury causes add 104,988 heat deaths per year, and all nine have negative mean cold values, summing to −118,825: the curves indicate lower injury risk on cold days than at the TMREL. These negative values stay in the cold total, so a reader adding only the positive causes will arrive at a larger number than ours.

Breakdowns by cause become much less certain as they get narrower. At world level under SSP2-4.5, the 95% interval includes zero for 7 of 17 causes for heat and 1 of 17 for cold. For a single cause in a single country, it includes zero in most cells for heat and about a third of cells for cold (60.2%

Table 6. Deaths per year by cause, SSP2-4.5, 2041–2050, 24 common models: world mean with 95% interval, and means for the high-income group and the low- and middle-income (LMIC) group.

	Heat			Cold		
Cause	World (95%)	High	LMIC	World (95%)	High	LMIC
Ischemic heart disease	168,892 (88,605 to 294,425)	15,116	153,776	562,182 (419,381 to 757,498)	140,555	421,627
Stroke	144,203 (76,211 to 236,264)	6,586	137,617	413,001 (325,157 to 509,460)	95,317	317,684
COPD	78,641 (14,117 to 162,583)	3,955	74,686	475,014 (369,004 to 593,297)	105,515	369,499
Lower respiratory infections	74,796 (25,270 to 136,866)	8,759	66,037	197,171 (156,005 to 237,994)	100,834	96,337
Diabetes	85,485 (47,878 to 135,175)	5,791	79,693	84,124 (59,836 to 110,368)	23,452	60,672
Chronic kidney disease	46,154 (2,538 to 100,577)	7,197	38,957	176,376 (134,927 to 219,950)	75,670	100,706
Hypertensive heart disease	26,790 (−878 to 64,360)	3,359	23,431	182,882 (133,129 to 242,940)	49,988	132,894
Cardiomyopathy and myocarditis	−442 (−13,018 to 12,000)	1,544	−1,986	27,009 (19,239 to 34,936)	13,731	13,278
Road injuries	49,986 (26,159 to 87,013)	4,017	45,969	−28,758 (−42,339 to −18,724)	−5,646	−23,112
Self-harm	21,798 (5,553 to 40,586)	1,429	20,369	−36,748 (−47,673 to −27,960)	−11,269	−25,479
Interpersonal violence	14,647 (8,425 to 23,176)	541	14,106	−11,125 (−15,197 to −7,879)	−2,321	−8,805
Drowning	7,545 (283 to 15,411)	516	7,029	−23,603 (−29,669 to −18,418)	−4,446	−19,157
Exposure to mechanical forces	3,161 (−146 to 6,960)	206	2,955	−4,723 (−6,256 to −3,121)	−940	−3,783
Exposure to forces of nature	2,761 (−1,775 to 12,144)	38	2,723	−1,759 (−3,999 to 739)	−293	−1,467
Animal contact	2,527 (−9,930 to 15,813)	23	2,503	−4,881 (−9,533 to −1,142)	−240	−4,640
Other transport injuries	1,728 (−2,785 to 6,147)	125	1,603	−3,610 (−4,990 to −2,361)	−1,196	−2,414
Other unintentional injuries	835 (−1,505 to 3,213)	59	776	−3,618 (−4,783 to −2,550)	−666	−2,952

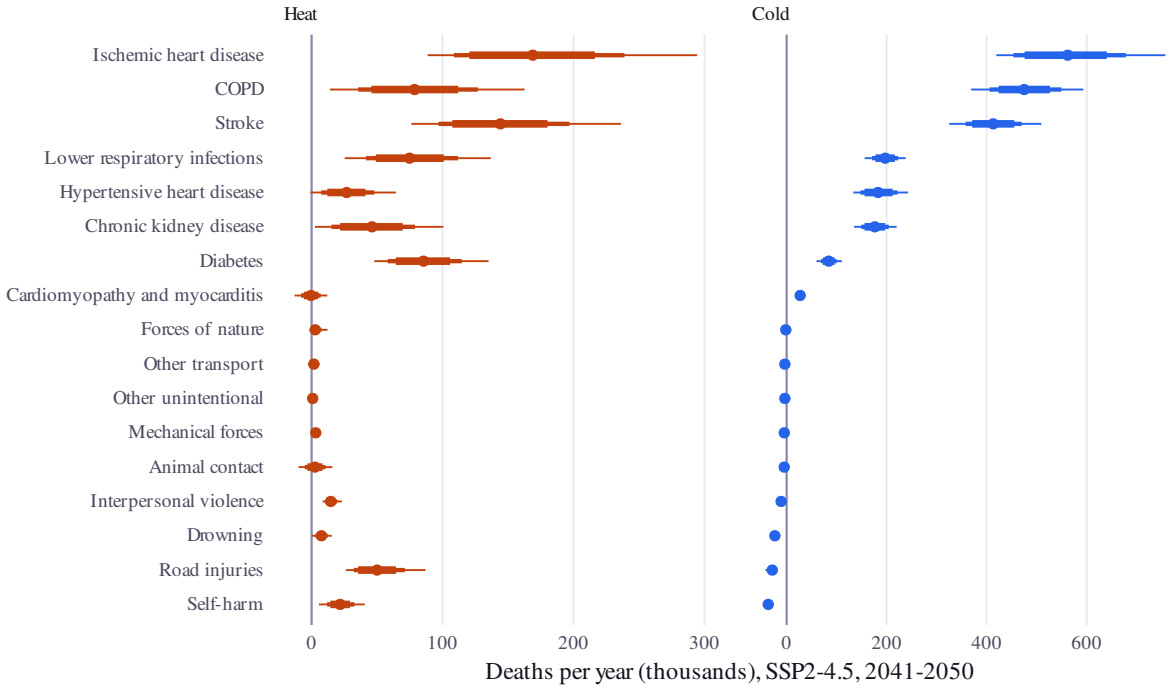


Figure 6. World deaths per year by cause, SSP2-4.5, 2041–2050, 24 common climate models. Thin, medium and thick bars are the 95%, 80% and 66% intervals. The horizontal scales differ between panels, and causes are ordered by mean cold deaths.

and 30.7% of 3,452 non-zero cells under SSP5-8.5). For that reason we report causes at world and income-group level here, and country by cause only in Appendix B, where cells whose interval includes zero are in parentheses. Plain country totals rarely have this problem: for deaths, the interval includes zero in at most 2 of 204 countries.

3.7 Which ages, and which sex?

Table 7 and Figure 7 show the three age groups the team asked for. People aged 65 and over account for a growing majority of attributable deaths: 59% of heat deaths in 2022–2030 and 72% in 2041–2050, and 88% and 93% of cold deaths. Children under 5 account for 4.9% of heat deaths in 2022–2030 and 1.5% in 2041–2050, falling in count as well (21,009 to 10,766 per year) with falling under-5 mortality in the forecast. Women aged 15 to 49 have 18,679 and 19,106 heat deaths per year in the two decades. Heat deaths at 15 to 49 are about three times higher in men than in women, which we suspect reflects the injury causes, although we have not split these slices by cause. For the same reason, we suspect, men aged 15 to 49 have a negative cold total at world level (−7,116 deaths per year in 2041–2050, with an interval from −13,182 to −725).

Table 7. World deaths per year and crude rate per 100,000 in that age and sex group, SSP2-4.5, 24 common models, mean and 95% interval. The slices do not sum to the total: ages 5–14 and 50–64 are not exported.

Period	Group	Heat deaths	Rate	Cold deaths	Rate
2022–2030	Under 5, male	11,092 (4,960 to 20,041)	3.4	3,562 (1,621 to 5,938)	1.1
	Under 5, female	9,917 (4,324 to 17,481)	3.3	3,673 (1,727 to 6,028)	1.2
	15–49, male	52,427 (39,689 to 69,712)	2.5	−4,526 (−10,164 to 725)	−0.2
	15–49, female	18,679 (13,958 to 25,003)	0.9	8,085 (6,040 to 10,422)	0.4
	65+, male	132,274 (98,226 to 179,887)	32.7	646,170 (575,613 to 719,151)	159.7
	65+, female	121,150 (88,670 to 165,302)	24.7	652,803 (544,919 to 738,074)	133.1
2041–2050	Under 5, male	5,843 (1,684 to 12,067)	1.9	972 (162 to 2,113)	0.3
	Under 5, female	4,923 (1,160 to 10,292)	1.7	1,059 (251 to 2,177)	0.4
	15–49, male	59,110 (40,119 to 89,142)	2.6	−7,116 (−13,182 to −725)	−0.3
	15–49, female	19,106 (11,687 to 30,639)	0.9	5,928 (3,350 to 9,598)	0.3
	65+, male	270,661 (188,786 to 395,895)	40.4	902,450 (750,280 to 1,064,557)	134.7
	65+, female	251,482 (166,278 to 379,694)	31.4	950,596 (728,685 to 1,170,536)	118.5

3.8 What about years of life lost?

Under SSP2-4.5, heat accounts for 12.8 million YLLs per year in 2041–2050 (95% interval 8.7 to 18.9 million) and cold for 23.8 million (18.9 to 29.4 million). Heat is a larger share of YLLs than of deaths (35% against 27%) because heat deaths include more young people. The ratio of SSP5-8.5 to SSP2-4.5 is 1.119 for heat YLLs and 0.976 for cold YLLs.

One feature needs care in any table of country YLLs. Because injury causes have negative cold values and fall at young ages, where remaining life expectancy is high, a country's cold YLLs can be negative alongside positive cold deaths: 21 countries in 2022–2030, 13 in 2031–2040 and 8 in 2041–2050 under SSP2-4.5, with the interval including zero in 69, 60 and 54. For example, in Burkina Faso under SSP1-2.6 in 2041–2050, cold deaths are +114.6 per year and cold YLLs −294.9. It follows that YLLs per death is not a usable statistic wherever signs mix (it reads 110.7 for men aged 15 to 49 at world level in 2022–2030). Where signs do not mix it behaves as expected: about 10 to 12 years at

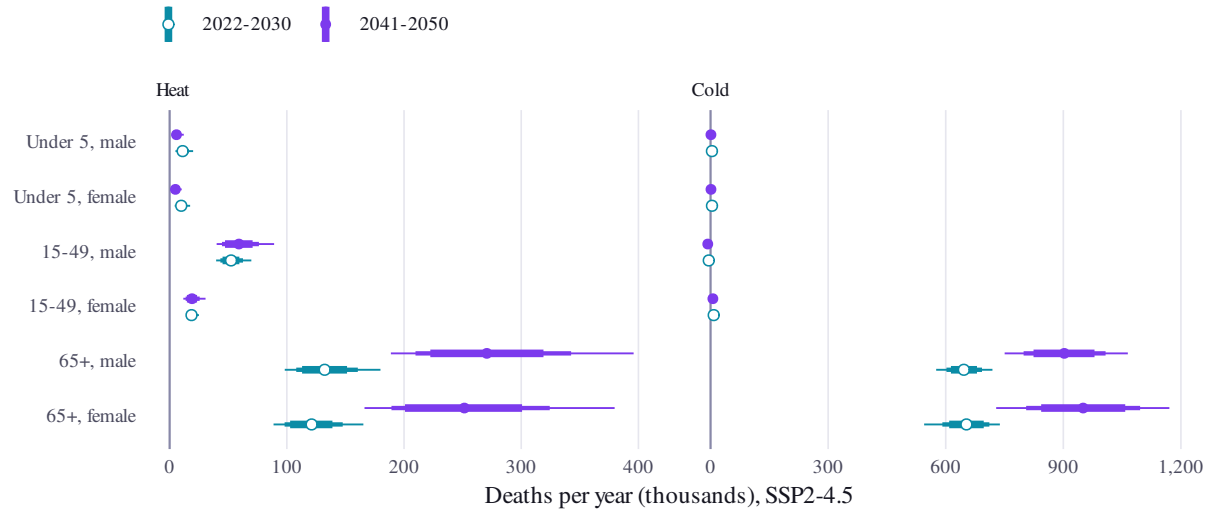


Figure 7. World deaths per year by age group and sex, SSP2-4.5, 2022–2030 and 2041–2050, 24 common climate models. Thin, medium and thick bars are the 95%, 80% and 66% intervals. The horizontal scales differ between panels.

65 and over, about 41 to 45 for heat at 15 to 49, and about 63 to 70 for heat under 5.

3.9 When does heat overtake cold?

Samuel asked for the year in which heat deaths overtake cold deaths in each country. To answer this we fitted, for each combination of climate model, draw and scenario, a straight line to yearly heat minus cold over 2022–2050, and took the crossing as the year in which the line reaches zero, recording “before 2022” if it is already above zero in 2022 and “after 2050” if it is still below zero in 2050. Table 8 counts countries by where the median of those 12,000 crossings falls, and this replaces our earlier first look, which used only model means. A straight line is a strong simplification, and we offer these as a summary of direction and not as estimates of a date.

Table 8. Countries (of 204) by where the median year heat overtakes cold falls, over 12,000 (model, draw) pairs, 24 common models.

Where the median falls	SSP1-2.6	SSP2-4.5	SSP3-7.0	SSP5-8.5
Heat above cold already in 2022	83	82	80	75
Median crossing within 2022–2050	6	8	11	16
Heat stays below cold through 2050	115	114	113	113

In most countries the answer is settled at one end. Heat is above cold from the start in India (in 85% of pairs) and Nigeria (100%), and stays below it through 2050 in China, the United States and the Russian Federation (100%). The year itself, however, cannot be pinned down in the countries where it falls within the period: under SSP2-4.5 there are eight (Table 9), and in all eight the 95% interval for the year runs from before 2022 to after 2050. We would therefore report for those countries the share of pairs in each category and not a year. Warming moves countries in the expected direction, with 16 countries in the middle category under SSP5-8.5, against 6 under SSP1-2.6, though the numbers are small.

Table 9. Countries whose median crossing falls within 2022–2050 under SSP2-4.5: median year and the share of (model, draw) pairs in each category.

Country	Median year	Before 2022	Within 2022–2050	After 2050
Tonga	2023	48%	40%	12%
Sao Tome and Principe	2027	37%	41%	23%
Iraq	2033	33%	36%	32%
Dominican Republic	2034	23%	46%	32%
Vanuatu	2036	16%	51%	33%
Lao People's Democratic Republic	2039	20%	45%	36%
Honduras	2040	30%	27%	43%
Niue	2042	12%	49%	39%

3.10 Do the results depend on which climate models are used?

The headline numbers use the 24 models common to all four scenarios. Table 10 compares two alternatives: paired sets, in which each target uses every model that has both its run and an SSP2-4.5 run (26, 24 and 27 models), and all available, in which the target uses its models and SSP2-4.5 uses all 27. The ratio moves by up to 0.7 points for heat under SSP5-8.5 (1.127, 1.134 and 1.134) and by a point for SSP3-7.0 under the all-available set (0.998 against 1.008), in which different models are being compared, and the conclusions above do not change. Models with no SSP2-4.5 run cannot be scaled and so are absent from all three sets; among them are HadGEM3-GC31-MM and TaiESM1, two of the three hottest of the 29 models under SSP5-8.5 in 2041–2050.

Table 10. World mean deaths per year for the target scenario, and ratio to SSP2-4.5, 2041–2050, under the three choices of climate models.

		Common 24		Paired		All available	
Measure	Scenario	Deaths	Ratio	Deaths	Ratio	Deaths	Ratio
Heat	SSP1-2.6	703,149	0.964	710,066	0.963	710,066	0.964
	SSP3-7.0	735,317	1.008	735,317	1.008	735,317	0.998
	SSP5-8.5	821,948	1.127	835,842	1.134	835,842	1.134
Cold	SSP1-2.6	2,005,660	1.003	2,002,463	1.004	2,002,463	1.004
	SSP3-7.0	2,006,773	1.004	2,006,773	1.004	2,006,773	1.006
	SSP5-8.5	1,950,759	0.976	1,944,173	0.974	1,944,173	0.974
Heat + cold	SSP1-2.6	2,708,809	0.993	2,712,529	0.993	2,712,529	0.993
	SSP3-7.0	2,742,090	1.005	2,742,090	1.005	2,742,090	1.004
	SSP5-8.5	2,772,707	1.016	2,780,015	1.018	2,780,015	1.018

4 How the numbers were checked

The per-draw files were compared with the summary tables Caspar's machine produced independently of them. For all 204 countries, the yearly means and quantiles match those tables to within the 6-digit rounding of the export, causes summed in a draw match the yearly totals, the 17 causes and three age groups are complete in every file, and under-5 YLLs per death lie within the life table's value for the right sex, apart from a small number of rows (90 of 162,344) that we attribute to deaths of mixed sign. Pooled numbers were then computed twice, by the production script and by independent code, and

these agree to within 10^{-10} in relative terms at world level and for all 253,368 country-by-cause and country-by-age cells of the three target scenarios. Before pooling we wrote down what we expected, and the world heat values matched to within 1 death in 1,000. Two expectations were wrong: we expected cold deaths to be about flat (they rise by 35%), and we expected stroke to rank second for cold (it is COPD). The full review is in `docs/reviews/workflow-b-draw-export-review-2026-11-05.org`. Against GBD 2023-round estimates for 2022, our earlier comparison on the SSP2-4.5 runs gave ratios of 0.68 to 0.77 for heat and 0.93 to 0.98 for cold (review of September 24, 2026), so our heat numbers run below that reference, and we have not resolved why.

5 Discussion

The main reading of these results is that most of the rise in temperature-attributable deaths to 2050 is common to all four scenarios, and that the scenarios differ, by 2041–2050, in the split between heat and cold more than in the total. That reading rests on scenarios whose temperatures have not diverged far by 2050 (SSP5-8.5 is 0.50°C warmer than SSP2-4.5 in 2050 in a death-weighted average over countries, from the review of October 26, 2026), and we would not extend it beyond 2050, when the scenarios diverge further.

Several assumptions limit what these numbers can say. First, the exposure-response curves do not change over time, so nothing here tests adaptation, whether through air conditioning, health systems or physiology. Second, we hold IHME's population and mortality forecasts fixed across scenarios, apart from the temperature scalar, so climate does not affect the other risks, and the forecasts' own features (the Gulf states, or event-scale disaster draws) pass straight through. Third, the intervals cover the curves, the mortality forecast's draws and disagreement between climate models, on 24 models that are not independent; they do not cover structural choices such as the bias correction of the temperature inputs, the 0.25-degree resolution, or the choice of curves. Fourth, our YLLs are not GBD's, and our rates are crude.

Open items remain. The SSP3-7.0 result in parts of Western and Central Europe (fewer heat deaths with annual mean temperature about unchanged) has not been explained, and would need summer temperature from Caspar's machine. A split of the rise by decade would also need age-by-year output from that machine. The income-group mapping needs a check by a second person, and the Foresight forecast's handling of migrant populations is something we would want to ask IHME about before leaning on the Gulf rates.

A Country totals

Table 11 lists every country under SSP2-4.5 for 2041–2050. The other scenarios, decades, measures and interval levels are in `tables/country_totals.csv` alongside this report. Ratios mean little where the deaths are few: Greenland's 3.01 rests on far below one heat death per year, so the deaths should be read beside every ratio.

Table 11. Country totals under SSP2-4.5, 2041–2050, on the 24 common models: mean deaths per year with 95% interval, crude rate per 100,000 people per year, and the SSP5-8.5 to SSP2-4.5 ratio of mean heat deaths. Income group: H high, UM upper middle, LM lower middle, L low, – not classified. An asterisk marks the two countries whose totals carry a disaster-forecast tail.

Country	Inc.	Heat deaths (95%)	Rate	Cold deaths (95%)	Rate	Net rate	Ratio
Afghanistan	L	1,704 (967 to 2,761)	2.7	8,718 (5,281 to 13,473)	13.9	16.6	1.09
Albania	UM	62 (20 to 134)	2.6	1,761 (1,358 to 2,210)	73.4	76.0	1.42
Algeria	UM	2,895 (1,642 to 4,755)	5.3	18,112 (13,192 to 24,417)	33.3	38.6	1.19
American Samoa	H	2.9 (0.9 to 6.3)	4.8	0.1 (0.0 to 0.5)	0.2	5.0	1.23
Andorra	H	0.0 (0.0 to 0.1)	0.0	37 (26 to 50)	48.7	48.7	1.85
Angola	LM	554 (236 to 1,089)	0.9	1,278 (788 to 1,803)	2.1	2.9	1.25
Antigua and Barbuda	H	3.5 (0.8 to 7.8)	3.8	0.7 (0.1 to 2.1)	0.8	4.6	1.27
Argentina	UM	1,203 (659 to 1,929)	2.5	16,752 (13,271 to 20,832)	35.1	37.6	1.21
Armenia	UM	106 (42 to 216)	4.3	1,529 (1,130 to 2,010)	61.8	66.1	1.13
Australia	H	298 (169 to 487)	0.9	7,412 (5,709 to 9,145)	21.9	22.8	1.17
Austria	H	104 (38 to 219)	1.1	4,110 (3,178 to 4,988)	41.8	42.8	1.15
Azerbaijan	UM	640 (328 to 1,087)	5.6	6,660 (4,844 to 9,002)	58.5	64.1	1.21
Bahamas	H	21 (8.2 to 41)	4.8	18 (7.0 to 31)	4.1	9.0	1.24
Bahrain	H	548 (285 to 902)	25.7	190 (72 to 315)	8.9	34.6	1.10
Bangladesh	LM	20,979 (10,195 to 35,270)	11.6	12,214 (6,759 to 19,001)	6.8	18.4	1.18
Barbados	H	13 (3.6 to 28)	4.8	0.9 (0.0 to 3.2)	0.3	5.1	1.27
Belarus	UM	158 (28 to 507)	1.9	4,844 (3,398 to 6,638)	57.9	59.8	1.21
Belgium	H	42 (10 to 107)	0.3	5,156 (3,970 to 6,269)	40.3	40.7	1.04
Belize	UM	43 (22 to 82)	6.9	9.9 (3.7 to 19)	1.6	8.5	1.25
Benin	LM	1,573 (773 to 2,708)	5.9	48 (0.7 to 178)	0.2	6.1	1.11
Bermuda	H	1.6 (0.1 to 3.7)	2.8	11 (6.9 to 16)	19.7	22.5	1.11
Bhutan	LM	3.4 (0.9 to 7.3)	0.4	218 (156 to 294)	25.3	25.7	1.05
Bolivia (Plurinational State of)	LM	394 (208 to 676)	2.6	3,048 (2,333 to 3,857)	20.1	22.7	1.23
Bosnia and Herzegovina	UM	56 (17 to 114)	2.3	1,957 (1,495 to 2,557)	79.4	81.6	1.19
Botswana	UM	155 (80 to 248)	5.2	296 (222 to 384)	9.9	15.0	1.14
Brazil	UM	9,262 (5,086 to 16,055)	3.7	18,740 (12,038 to 27,359)	7.5	11.3	1.22
Brunei Darussalam	H	31 (7.8 to 72)	6.7	0.7 (0.0 to 2.6)	0.2	6.8	1.41
Bulgaria	H	168 (62 to 327)	3.3	5,505 (4,397 to 6,611)	107.1	110.3	1.14
Burkina Faso	L	5,536 (2,887 to 9,045)	11.8	105 (−1.5 to 398)	0.2	12.0	1.07
Burundi	L	5.1 (0.6 to 12)	0.0	895 (544 to 1,326)	3.8	3.9	1.46
Cabo Verde	UM	5.4 (1.5 to 12)	0.9	24 (14 to 39)	3.9	4.7	1.18
Cambodia	LM	4,519 (2,430 to 7,349)	21.3	97 (14 to 271)	0.5	21.8	1.14
Cameroon	LM	2,172 (1,218 to 3,465)	4.1	767 (447 to 1,168)	1.5	5.6	1.08
Canada	H	549 (247 to 1,147)	1.2	15,062 (11,774 to 17,992)	31.7	32.9	1.14
Central African Republic	L	517 (253 to 956)	7.4	30 (6.2 to 72)	0.4	7.8	1.11
Chad	L	4,390 (2,115 to 7,401)	9.7	127 (14 to 448)	0.3	10.0	1.04
Chile	H	5.3 (0.8 to 15)	0.0	7,686 (6,230 to 9,169)	35.7	35.8	1.87
China	UM	113,009 (70,115 to 190,239)	8.8	792,478 (621,344 to 978,559)	61.4	70.2	1.14
Colombia	UM	1,712 (776 to 3,112)	3.1	8,336 (5,947 to 11,297)	15.0	18.1	1.30
Comoros	LM	16 (6.4 to 32)	1.8	5.9 (1.8 to 12)	0.7	2.5	1.28
Congo	LM	181 (55 to 444)	2.4	69 (32 to 120)	0.9	3.3	1.34
Cook Islands	–	0.2 (0.0 to 0.6)	1.0	1.0 (0.4 to 2.2)	5.8	6.8	1.41
Costa Rica	H	30 (14 to 61)	0.6	525 (339 to 762)	10.3	10.9	1.25
Croatia	H	83 (29 to 164)	2.6	2,379 (1,840 to 3,033)	75.0	77.6	1.20
Cuba	UM	991 (538 to 1,817)	10.1	530 (269 to 819)	5.4	15.5	1.24
Cyprus	H	142 (75 to 236)	9.4	395 (292 to 504)	25.9	35.3	1.09
Czechia	H	138 (40 to 330)	1.4	5,418 (4,121 to 6,713)	55.1	56.5	1.19
Côte d'Ivoire	LM	1,942 (954 to 3,477)	4.2	128 (12 to 375)	0.3	4.5	1.23
Democratic People's Republic of Korea	L	1,227 (556 to 2,287)	4.9	20,629 (15,753 to 26,084)	82.6	87.5	1.02
Democratic Republic of the Congo	L	2,369 (809 to 5,464)	1.6	2,243 (1,249 to 3,471)	1.5	3.1	1.27
Denmark	H	6.8 (0.6 to 29)	0.1	2,801 (2,203 to 3,379)	43.3	43.5	1.20
Djibouti	LM	363 (190 to 581)	19.2	29 (2.1 to 76)	1.5	20.7	1.06
Dominica	UM	1.4 (0.3 to 3.6)	2.2	0.5 (0.1 to 1.3)	0.8	3.0	1.38
Dominican Republic	UM	418 (167 to 822)	3.3	316 (177 to 489)	2.5	5.8	1.33
Ecuador	UM	158 (44 to 357)	0.7	3,525 (2,708 to 4,410)	15.7	16.4	1.42
Egypt	LM	14,298 (7,576 to 22,910)	9.3	30,965 (22,705 to 40,882)	20.1	29.4	1.12

Country	Inc.	Heat deaths (95%)	Rate	Cold deaths (95%)	Rate	Net rate	Ratio
El Salvador	UM	527 (252 to 1,100)	9.3	50 (14 to 115)	0.9	10.2	1.36
Equatorial Guinea	UM	25 (8.1 to 57)	0.9	7.9 (2.0 to 20)	0.3	1.1	1.34
Eritrea	L	714 (379 to 1,202)	7.9	296 (157 to 497)	3.3	11.1	1.09
Estonia	H	4.1 (0.6 to 13)	0.3	662 (450 to 909)	53.6	53.9	1.25
Eswatini	LM	25 (10 to 44)	1.9	152 (113 to 198)	11.6	13.5	1.17
Ethiopia	L	1,696 (978 to 2,650)	0.9	8,330 (5,790 to 11,351)	4.3	5.1	1.09
Fiji	UM	19 (6.0 to 41)	2.1	43 (24 to 84)	4.8	6.8	1.41
Finland	H	11 (1.2 to 37)	0.2	2,061 (1,426 to 2,647)	37.6	37.8	1.27
France	H	305 (133 to 574)	0.5	24,651 (19,421 to 29,781)	37.5	37.9	1.20
Gabon	UM	54 (21 to 115)	2.0	14 (5.9 to 27)	0.5	2.5	1.31
Gambia	L	343 (187 to 546)	9.6	16 (4.8 to 35)	0.5	10.1	1.11
Georgia	UM	79 (37 to 145)	2.3	2,131 (1,627 to 2,701)	63.2	65.5	1.21
Germany	H	713 (244 to 1,578)	0.8	41,058 (31,504 to 49,247)	47.5	48.3	1.12
Ghana	LM	3,890 (2,013 to 6,716)	7.6	210 (14 to 615)	0.4	8.0	1.12
Greece	H	567 (211 to 1,164)	6.8	4,842 (3,677 to 6,021)	58.1	64.9	1.29
Greenland	H	0.0 (0.0 to 0.0)	0.0	27 (18 to 39)	45.9	45.9	3.01
Grenada	UM	4.2 (1.0 to 9.8)	4.4	0.2 (0.0 to 1.0)	0.2	4.6	1.26
Guam	H	6.8 (1.2 to 17)	4.4	0.3 (0.0 to 1.1)	0.2	4.6	1.44
Guatemala	UM	256 (137 to 496)	1.3	1,921 (1,309 to 2,634)	10.1	11.4	1.25
Guinea	LM	974 (554 to 1,577)	4.0	109 (27 to 240)	0.4	4.4	1.18
Guinea-Bissau	L	431 (233 to 692)	12.2	3.4 (0.0 to 15)	0.1	12.3	1.13
Guyana	H	49 (17 to 110)	7.8	2.0 (0.3 to 5.9)	0.3	8.1	1.40
Haiti*	LM	1,615 (212 to 7,540)	10.0	416 (101 to 1,934)	2.6	12.5	1.04
Honduras*	LM	389 (178 to 802)	2.9	429 (219 to 703)	3.2	6.2	1.27
Hungary	H	268 (86 to 589)	2.9	5,715 (4,274 to 7,421)	61.7	64.6	1.08
Iceland	H	0.0 (0.0 to 0.0)	0.0	132 (89 to 179)	27.8	27.8	1.18
India	LM	231,555 (146,386 to 359,901)	14.6	111,580 (60,560 to 167,559)	7.0	21.6	1.10
Indonesia	UM	16,286 (6,248 to 32,378)	5.2	4,755 (2,672 to 7,625)	1.5	6.7	1.31
Iran (Islamic Republic of)	UM	7,210 (4,753 to 10,577)	7.3	33,421 (24,127 to 44,807)	33.8	41.0	1.11
Iraq	UM	8,837 (5,022 to 13,714)	15.0	8,508 (4,920 to 12,845)	14.5	29.5	1.10
Ireland	H	0.6 (0.0 to 3.2)	0.0	2,197 (1,690 to 2,668)	36.1	36.1	1.25
Israel	H	529 (222 to 964)	4.1	1,354 (937 to 1,755)	10.6	14.8	1.12
Italy	H	886 (333 to 1,808)	1.7	29,582 (22,157 to 35,796)	57.0	58.7	1.45
Jamaica	UM	135 (63 to 260)	5.1	29 (9.4 to 58)	1.1	6.2	1.28
Japan	H	3,996 (1,868 to 8,531)	3.6	46,620 (33,100 to 58,096)	41.6	45.2	1.20
Jordan	UM	698 (308 to 1,278)	3.3	3,050 (2,178 to 4,231)	14.4	17.7	1.21
Kazakhstan	UM	709 (436 to 1,099)	3.1	8,988 (6,839 to 11,431)	39.3	42.4	1.09
Kenya	LM	659 (365 to 1,081)	0.9	4,108 (3,037 to 5,338)	5.8	6.7	1.11
Kiribati	LM	7.6 (2.8 to 15)	5.3	0.3 (0.0 to 1.1)	0.2	5.6	1.34
Kuwait	H	1,210 (669 to 1,940)	17.8	493 (211 to 841)	7.3	25.1	1.08
Kyrgyzstan	LM	101 (39 to 204)	1.2	2,857 (2,011 to 3,870)	34.5	35.8	1.14
Lao People's Democratic Republic	LM	582 (324 to 1,013)	6.2	527 (352 to 734)	5.6	11.9	1.27
Latvia	H	12 (1.5 to 46)	0.9	959 (718 to 1,201)	67.7	68.6	1.30
Lebanon	LM	71 (24 to 158)	1.2	1,954 (1,383 to 2,648)	33.4	34.6	1.42
Lesotho	LM	0.4 (0.0 to 1.2)	0.0	784 (663 to 940)	35.5	35.5	1.52
Liberia	L	153 (49 to 310)	1.8	22 (4.4 to 56)	0.3	2.0	1.39
Libya	UM	837 (396 to 1,497)	11.9	2,030 (1,325 to 2,943)	28.9	40.7	1.04
Lithuania	H	25 (3.6 to 89)	1.2	1,276 (903 to 1,657)	63.2	64.5	1.26
Luxembourg	H	4.1 (1.3 to 9.8)	0.4	269 (219 to 323)	28.6	29.0	1.10
Madagascar	L	397 (207 to 649)	0.9	3,428 (2,588 to 4,375)	7.4	8.2	1.27
Malawi	L	365 (184 to 637)	1.2	944 (639 to 1,351)	3.1	4.3	1.24
Malaysia	UM	1,941 (528 to 4,416)	5.1	109 (26 to 272)	0.3	5.4	1.33
Maldives	UM	22 (4.3 to 55)	2.9	0.6 (0.0 to 3.1)	0.1	3.0	1.42
Mali	L	5,248 (2,983 to 8,194)	10.0	172 (1.0 to 491)	0.3	10.3	1.08
Malta	H	21 (4.3 to 47)	3.9	123 (83 to 166)	22.9	26.9	1.19
Marshall Islands	UM	4.8 (1.3 to 12)	6.8	0.0 (0.0 to 0.1)	0.0	6.8	1.40
Mauritania	LM	1,225 (687 to 1,892)	17.0	68 (21 to 136)	0.9	17.9	1.04
Mauritius	UM	14 (0.5 to 44)	1.3	121 (63 to 224)	10.5	11.8	1.39
Mexico	UM	6,024 (4,110 to 8,652)	4.1	33,817 (25,427 to 43,558)	23.0	27.1	1.11
Micronesia (Federated States of)	UM	9.5 (2.1 to 23)	9.3	0.0 (0.0 to 0.2)	0.0	9.3	1.38
Monaco	H	0.1 (0.0 to 0.3)	0.3	18 (13 to 23)	57.4	57.7	2.04

Country	Inc.	Heat deaths (95%)	Rate	Cold deaths (95%)	Rate	Net rate	Ratio
Mongolia	UM	25 (8.7 to 61)	0.5	963 (536 to 1,397)	18.8	19.3	1.13
Montenegro	UM	8.4 (2.7 to 18)	1.6	380 (307 to 467)	70.0	71.6	1.37
Morocco	LM	1,966 (951 to 3,541)	4.9	16,376 (11,981 to 22,379)	40.8	45.6	1.16
Mozambique	L	1,383 (773 to 2,223)	2.6	666 (410 to 984)	1.2	3.8	1.21
Myanmar	LM	10,658 (6,056 to 17,135)	16.7	3,399 (1,832 to 5,275)	5.3	22.0	1.19
Namibia	LM	130 (56 to 224)	3.9	227 (164 to 300)	6.8	10.7	1.22
Nauru	H	1.6 (0.5 to 3.6)	10.5	0.0 (0.0 to 0.0)	0.0	10.5	1.29
Nepal	LM	1,624 (864 to 2,662)	4.8	5,422 (3,778 to 7,565)	16.2	21.0	1.13
Netherlands	H	44 (8.9 to 125)	0.2	7,858 (6,102 to 9,554)	42.9	43.1	1.03
New Zealand	H	0.2 (0.0 to 1.1)	0.0	2,025 (1,566 to 2,483)	30.1	30.1	1.04
Nicaragua	LM	693 (325 to 1,287)	8.9	80 (31 to 172)	1.0	10.0	1.30
Niger	L	7,289 (3,591 to 11,848)	11.5	317 (1.8 to 926)	0.5	12.0	1.06
Nigeria	LM	23,046 (13,388 to 36,330)	5.6	2,022 (505 to 4,517)	0.5	6.1	1.09
Niue	–	0.1 (0.0 to 0.1)	3.0	0.0 (0.0 to 0.1)	2.6	5.7	1.30
North Macedonia	UM	57 (20 to 105)	3.1	1,572 (1,229 to 1,977)	85.1	88.2	1.19
Northern Mariana Islands	H	4.8 (1.3 to 10)	9.3	0.4 (0.0 to 1.1)	0.7	10.0	1.33
Norway	H	3.6 (0.6 to 11)	0.1	2,040 (1,580 to 2,444)	32.8	32.8	1.30
Oman	H	1,683 (847 to 2,684)	24.6	408 (150 to 740)	6.0	30.6	1.04
Pakistan	LM	38,826 (24,065 to 60,678)	13.2	29,405 (18,577 to 42,233)	10.0	23.3	1.08
Palau	H	2.0 (0.4 to 4.8)	12.0	0.0 (0.0 to 0.1)	0.1	12.0	1.39
Palestine	LM	363 (177 to 615)	5.2	1,042 (759 to 1,399)	14.9	20.2	1.08
Panama	H	155 (60 to 320)	2.7	30 (16 to 47)	0.5	3.2	1.37
Papua New Guinea	LM	210 (80 to 445)	1.1	1,545 (1,084 to 2,108)	7.8	8.9	1.33
Paraguay	UM	503 (208 to 899)	6.3	874 (556 to 1,308)	10.9	17.2	1.14
Peru	UM	255 (112 to 502)	0.6	6,962 (5,465 to 8,679)	16.7	17.3	1.25
Philippines	UM	10,123 (4,784 to 18,661)	6.9	1,153 (512 to 2,168)	0.8	7.7	1.30
Poland	H	450 (109 to 1,116)	1.3	18,957 (14,911 to 23,038)	56.6	57.9	1.16
Portugal	H	46 (13 to 95)	0.4	5,262 (3,969 to 6,535)	50.0	50.5	1.32
Puerto Rico	H	110 (37 to 231)	4.4	50 (15 to 99)	2.0	6.4	1.29
Qatar	H	1,209 (607 to 1,948)	18.9	268 (72 to 497)	4.2	23.1	1.03
Republic of Korea	H	1,344 (624 to 2,532)	2.7	24,725 (18,930 to 30,047)	50.5	53.2	1.12
Republic of Moldova	UM	146 (41 to 346)	5.9	2,078 (1,408 to 2,940)	84.3	90.2	1.02
Romania	H	555 (232 to 1,025)	3.6	12,845 (9,965 to 16,152)	82.3	85.8	1.13
Russian Federation	H	2,664 (1,208 to 5,658)	1.9	70,183 (50,214 to 90,702)	51.0	52.9	1.16
Rwanda	L	0.4 (0.0 to 1.3)	0.0	1,421 (967 to 1,928)	6.1	6.1	1.74
Saint Kitts and Nevis	H	2.2 (0.6 to 5.0)	4.1	0.5 (0.1 to 1.3)	0.9	5.0	1.26
Saint Lucia	UM	11 (3.7 to 25)	6.4	0.7 (0.0 to 2.3)	0.4	6.7	1.26
Saint Vincent and the Grenadines	UM	4.8 (1.2 to 10)	4.8	0.3 (0.0 to 1.2)	0.3	5.2	1.29
Samoa	UM	4.7 (1.3 to 11)	1.6	0.7 (0.1 to 1.9)	0.3	1.8	1.25
San Marino	H	0.4 (0.1 to 1.2)	1.5	13 (8.3 to 17)	41.5	42.9	1.57
Sao Tome and Principe	LM	3.3 (0.9 to 7.4)	1.4	2.1 (0.6 to 4.3)	0.9	2.2	1.30
Saudi Arabia	H	15,247 (8,476 to 24,694)	33.4	5,915 (3,340 to 9,203)	13.0	46.4	1.07
Senegal	LM	2,233 (1,314 to 3,399)	8.7	168 (79 to 297)	0.7	9.4	1.05
Serbia	UM	224 (77 to 433)	3.1	5,559 (4,400 to 6,915)	76.2	79.2	1.13
Seychelles	H	3.8 (0.8 to 9.1)	2.8	1.1 (0.0 to 3.4)	0.8	3.7	1.32
Sierra Leone	L	546 (260 to 930)	3.7	56 (8.1 to 151)	0.4	4.1	1.20
Singapore	H	196 (3.3 to 942)	2.7	67 (0.0 to 318)	0.9	3.6	1.27
Slovakia	H	99 (30 to 214)	1.9	3,072 (2,401 to 3,774)	60.4	62.3	1.14
Slovenia	H	23 (8.0 to 47)	1.1	983 (753 to 1,233)	46.4	47.5	1.19
Solomon Islands	LM	25 (9.5 to 51)	2.6	1.2 (0.2 to 3.0)	0.1	2.7	1.20
Somalia	L	3,697 (1,615 to 7,045)	8.9	69 (20 to 164)	0.2	9.0	1.14
South Africa	UM	1,048 (453 to 1,909)	1.6	14,873 (11,838 to 18,491)	22.8	24.4	1.18
South Sudan	L	2,275 (1,185 to 3,798)	11.4	33 (2.0 to 137)	0.2	11.6	1.06
Spain	H	1,034 (503 to 1,873)	2.0	20,026 (15,381 to 23,996)	39.3	41.3	1.31
Sri Lanka	UM	2,548 (1,396 to 4,252)	11.9	491 (247 to 827)	2.3	14.2	1.18
Sudan	L	12,995 (6,703 to 22,146)	20.2	1,540 (658 to 2,785)	2.4	22.6	1.05
Suriname	UM	60 (18 to 151)	9.1	1.2 (0.0 to 5.2)	0.2	9.3	1.33
Sweden	H	10 (1.5 to 39)	0.1	3,753 (2,803 to 4,474)	29.0	29.0	1.37
Switzerland	H	45 (14 to 110)	0.5	3,650 (2,742 to 4,397)	36.3	36.8	1.22
Syrian Arab Republic	L	2,228 (1,157 to 3,731)	13.6	7,293 (5,165 to 9,989)	44.5	58.1	1.08
Taiwan	H	1,876 (985 to 3,080)	8.8	4,159 (3,084 to 5,317)	19.4	28.2	1.12

Country	Inc.	Heat deaths (95%)	Rate	Cold deaths (95%)	Rate	Net rate	Ratio
Tajikistan	LM	440 (239 to 763)	3.0	4,422 (3,202 to 5,874)	30.1	33.1	1.16
Thailand	UM	16,621 (8,823 to 27,752)	27.7	1,351 (364 to 2,799)	2.3	30.0	1.18
Timor-Leste	LM	14 (5.3 to 30)	0.7	27 (13 to 48)	1.4	2.1	1.33
Togo	LM	950 (483 to 1,645)	7.5	51 (4.1 to 157)	0.4	7.9	1.11
Tokelau	–	0.1 (0.0 to 0.3)	8.1	0.0 (0.0 to 0.0)	0.0	8.2	1.34
Tonga	UM	2.3 (0.8 to 4.8)	1.8	1.1 (0.5 to 2.1)	0.9	2.6	1.26
Trinidad and Tobago	H	86 (33 to 172)	7.8	3.2 (0.2 to 9.2)	0.3	8.1	1.29
Tunisia	LM	1,205 (623 to 1,946)	10.0	4,577 (3,408 to 6,045)	37.8	47.8	1.11
Turkmenistan	UM	812 (494 to 1,274)	12.6	2,435 (1,667 to 3,435)	37.9	50.5	1.08
Tuvalu	UM	1.5 (0.4 to 3.2)	9.1	0.0 (0.0 to 0.0)	0.0	9.1	1.33
Türkiye	UM	2,702 (1,617 to 4,200)	3.2	41,641 (32,137 to 53,192)	48.9	52.1	1.18
Uganda	L	264 (116 to 487)	0.3	843 (464 to 1,317)	1.0	1.3	1.12
Ukraine	UM	2,538 (785 to 5,538)	7.6	30,437 (21,938 to 41,040)	91.7	99.3	1.10
United Arab Emirates	H	5,390 (2,665 to 8,769)	29.2	1,324 (420 to 2,587)	7.2	36.4	1.03
United Kingdom	H	44 (3.0 to 168)	0.1	31,362 (25,150 to 36,534)	42.9	42.9	1.15
United Republic of Tanzania	LM	654 (335 to 1,132)	0.7	2,031 (1,348 to 2,960)	2.1	2.8	1.25
United States Virgin Islands	H	5.1 (1.6 to 10.0)	7.4	1.0 (0.2 to 2.6)	1.5	8.9	1.27
United States of America	H	16,115 (10,568 to 24,281)	4.5	139,239 (109,884 to 164,847)	39.2	43.8	1.09
Uruguay	H	19 (6.2 to 40)	0.6	1,077 (821 to 1,350)	33.4	34.0	1.40
Uzbekistan	LM	3,328 (1,771 to 5,883)	7.7	18,657 (12,176 to 29,168)	42.9	50.5	1.13
Vanuatu	LM	7.9 (3.1 to 16)	1.7	6.2 (3.2 to 11)	1.3	3.0	1.33
Venezuela (Bolivarian Republic of)	LM	4,765 (2,067 to 10,078)	19.3	755 (217 to 1,887)	3.1	22.4	1.24
Viet Nam	UM	18,930 (11,276 to 29,695)	16.9	8,525 (4,992 to 13,045)	7.6	24.5	1.17
Yemen	L	3,141 (1,630 to 5,342)	6.0	3,659 (2,192 to 5,773)	7.0	13.0	1.09
Zambia	LM	425 (212 to 736)	1.2	978 (592 to 1,474)	2.9	4.1	1.17
Zimbabwe	LM	483 (237 to 812)	2.2	1,620 (1,177 to 2,161)	7.5	9.8	1.16

B Country by cause

Tables 12 and 13 give means for SSP2-4.5, 2041–2050. Other scenarios, decades and all interval levels are in the supplementary file `country_cause.csv.gz`, which is too large to carry here.

Table 12. Heat deaths per year by country and cause, SSP2-4.5, 2041–2050, mean over the 24 common models. Parentheses mark cells whose 95% interval includes zero. IHD ischemic heart disease; Str stroke; COPD chronic obstructive pulmonary disease; LRI lower respiratory infections; Diab diabetes; CKD chronic kidney disease; HHD hypertensive heart disease; Card cardiomyopathy and myocarditis; Road road injuries; Self self-harm; Viol interpersonal violence; Drwn drowning; Mech mechanical forces; Natr forces of nature; Anim animal contact; OTr other transport; OUn other unintentional.

Country	IHD	Str	COPD	LRI	Diab	CKD	HHD	Card	Road	Self	Viol	Drwn	Mech	Natr	Anim	OTr	OUn
Afghanistan	476	253	(45)	265	123	117	72	24	53	25	58	56	39	27	5.1	6.2	62
Albania	24	(23)	(0.9)	(4.4)	1.3	(1.1)	(0.7)	5.2	0.2	0.4	0.2	0.4	0.3	0.0	0.0	0.2	0.2
Algeria	926	570	(112)	375	232	304	(237)	47	36	16	2.8	12	(9.0)	4.6	1.5	4.5	6.6
American Samoa	(0.2)	(0.5)	(0.2)	(−0.2)	1.2	(0.5)	(0.0)	(−0.1)	0.2	0.1	0.1	(0.1)	(0.0)	(0.0)	(0.0)	(0.0)	(0.0)
Andorra	(0.0)	(0.0)	(0.0)	(0.0)	(0.0)	(0.0)	(0.0)	(0.0)	(0.0)	(0.0)	0.0	(0.0)	(0.0)	0.0	(0.0)	(0.0)	(0.0)
Angola	(8.0)	(73)	(3.0)	(66)	77	(25)	(−4.4)	(−23)	192	64	21	36	4.2	(0.4)	(6.3)	(3.4)	(2.3)
Antigua and Barbuda	(0.3)	(0.7)	(0.1)	(−0.2)	1.5	(0.7)	(0.2)	(−0.2)	0.1	0.0	0.1	(0.1)	(0.1)	(0.1)	(0.0)	(0.0)	(0.0)
Argentina	(71)	(83)	(21)	718	67	(74)	(7.4)	110	6.8	25	6.8	5.3	2.0	0.2	0.6	1.5	3.9
Armenia	70	(9.7)	(5.8)	(4.3)	(4.8)	(2.2)	(3.6)	(1.4)	(1.6)	0.8	0.3	0.7	(0.3)	(0.0)	0.1	(0.1)	0.4
Australia	62	45	(28)	61	33	(33)	(2.4)	12	3.7	15	0.8	1.7	0.4	0.1	0.3	0.5	0.3
Austria	51	(7.0)	(9.0)	(1.1)	(7.3)	(11)	(6.0)	(5.9)	(1.1)	2.9	0.1	0.5	(0.4)	(0.1)	0.1	0.3	0.1
Azerbaijan	371	(74)	(15)	39	35	13	(23)	56	1.9	(2.7)	(2.0)	5.3	0.4	0.2	0.1	0.3	0.7
Bahamas	(2.4)	2.8	(0.2)	(2.7)	4.2	(2.8)	(0.9)	(−0.3)	1.4	0.3	2.2	0.7	(0.0)	(0.1)	(0.1)	(0.1)	(0.0)
Bahrain	97	50	(27)	(31)	239	(50)	(22)	(−2.9)	27	(5.0)	2.4	(0.6)	(0.7)	0.0	(0.0)	(1.0)	(0.1)
Bangladesh	4,332	7,178	(1,621)	(1,304)	3,805	(842)	(574)	(−168)	504	277	154	204	(8.1)	(229)	(51)	(55)	(8.8)
Barbados	(0.9)	(2.9)	(0.3)	(−1.3)	7.1	(2.0)	(0.1)	(−0.9)	0.8	0.3	0.7	(0.2)	(0.0)	(0.0)	(0.0)	(0.0)	(0.0)
Belarus	138	(2.4)	(0.3)	(−0.9)	(0.5)	(−0.2)	(0.0)	(0.8)	(3.0)	(6.5)	1.2	3.6	(0.8)	(0.0)	0.1	0.2	0.9
Belgium	(14)	(5.7)	(7.5)	(4.6)	(1.9)	(3.6)	(0.3)	(1.2)	(0.7)	1.7	(0.1)	0.3	(0.1)	(0.0)	(0.0)	(0.1)	0.1
Belize	(4.4)	4.8	(1.4)	(0.4)	11	(6.6)	(1.2)	(−1.2)	5.8	1.5	5.6	(0.8)	(0.5)	(0.1)	(0.1)	(0.2)	(0.0)
Benin	(116)	352	(27)	(167)	199	(169)	(39)	(−44)	297	(92)	120	(19)	(17)	(0.0)	(0.6)	(4.3)	(0.4)
Bermuda	(0.2)	(0.2)	(0.0)	(0.5)	(0.2)	(0.2)	(0.0)	(0.1)	(0.0)	0.0	(0.0)	(0.0)	(0.0)	(0.0)	(0.0)	(0.0)	(0.0)
Bhutan	(0.3)	(0.7)	(0.6)	(0.5)	(0.4)	(0.2)	(0.0)	(0.2)	(0.0)	0.1	(0.0)	0.1	(0.1)	(0.0)	0.0	(0.0)	(0.0)
Bolivia (Plurinational State of)	48	50	(12)	76	69	67	(4.6)	(−1.0)	37	11	7.8	5.3	2.0	(0.4)	(0.5)	(2.0)	(1.6)
Bosnia and Herzegovina	30	(9.4)	(1.5)	(−0.2)	(6.9)	(0.7)	(1.0)	(4.5)	(0.3)	0.8	0.0	0.1	(0.0)	0.0	0.0	0.1	0.0
Botswana	(9.2)	(20)	(3.2)	63	25	(8.0)	(1.3)	8.6	(3.6)	5.4	2.7	2.7	(0.5)	0.0	1.6	(0.1)	0.4
Brazil	(1,096)	1,387	(548)	(1,418)	1,697	(740)	(167)	(−191)	883	319	1,035	90	(30)	(2.4)	(12)	(13)	(17)
Brunei Darussalam	(4.6)	(5.9)	(4.3)	(−3.1)	10	(5.8)	(0.8)	(−1.9)	2.9	0.7	0.2	(0.4)	(0.1)	(0.0)	(0.0)	(0.1)	(0.0)
Bulgaria	72	(29)	(2.6)	(2.0)	(7.3)	(3.7)	(41)	(3.7)	(1.3)	2.4	0.3	1.2	0.4	0.0	0.2	0.5	0.4
Burkina Faso	(431)	(681)	(61)	(1,519)	525	(307)	(215)	(−132)	1,088	(261)	487	(46)	(31)	(0.0)	(0.6)	(15)	(0.4)
Burundi	(−1.0)	(0.6)	(−0.4)	2.2	(0.7)	(0.0)	−0.6	(0.0)	1.1	0.9	0.0	0.7	0.1	0.0	0.4	(0.1)	0.2
Cabo Verde	(−0.5)	(0.7)	(−0.2)	2.4	(0.9)	(0.1)	(−0.5)	(0.0)	0.2	0.9	0.5	0.7	(0.0)	0.0	0.1	(0.0)	0.1
Cambodia	782	1,440	(263)	(352)	571	(283)	(234)	(−44)	435	(84)	57	(32)	(19)	(−0.3)	(−0.5)	(12)	(−0.1)
Cameroon	(191)	389	(33)	(340)	334	(220)	(59)	(−43)	431	119	39	(28)	(15)	(0.1)	(6.1)	(7.4)	(2.5)
Canada	200	(42)	(96)	(52)	(39)	(57)	(11)	(8.2)	11	22	2.0	3.4	(1.6)	0.1	0.3	3.5	0.5
Central African Republic	(23)	66	(9.5)	(67)	72	(16)	(13)	(−16)	176	36	28	(15)	(5.3)	(0.0)	(2.3)	(2.5)	(1.0)
Chad	(289)	(715)	(55)	(1,482)	(322)	(135)	(111)	(−74)	680	(193)	357	(62)	(50)	(0.1)	(1.6)	(10)	(1.5)
Chile	(0.8)	(0.5)	(−0.7)	(2.2)	(0.7)	(0.2)	(−0.1)	0.4	(0.2)	(0.6)	(0.1)	0.2	0.1	0.0	0.0	(0.0)	0.0
China	22,856	35,449	22,742	10,556	4,548	3,966	5,924	1,824	853	1,841	80	1,760	241	82	89	95	101
Colombia	522	186	(258)	(−34)	176	(162)	(43)	(−28)	174	45	196	(4.0)	(5.1)	(0.2)	(0.2)	(2.5)	(0.1)
Comoros	(−0.3)	(2.7)	(0.0)	(0.7)	3.6	(1.3)	(−0.9)	(−0.4)	4.1	1.4	1.8	0.8	(0.2)	(0.0)	(0.5)	(0.1)	0.1
Congo	(6.4)	28	(2.9)	(5.2)	37	(15)	(−1.9)	(−8.8)	57	20	8.2	6.8	(1.3)	(0.0)	(2.0)	(1.3)	(0.4)
Cook Islands	(0.0)	(0.0)	(0.0)	(0.0)	(0.1)	(0.0)	(0.0)	(0.0)	0.0	0.0	0.0	0.0	(0.0)	(0.0)	(0.0)	(0.0)	0.0
Costa Rica	(3.3)	3.3	(2.8)	(−0.6)	5.7	(5.5)	(0.4)	−1.2	5.8	2.2	2.5	(0.4)	(0.1)	(0.0)	(0.0)	(0.1)	(0.0)
Côte d'Ivoire	(192)	384	(38)	(86)	320	(184)	(59)	(−80)	340	157	200	(26)	(20)	(0.1)	(6.0)	(6.0)	(2.4)

Country	IHD	Str	COPD	LRI	Diab	CKD	HHD	Card	Road	Self	Viol	Drwn	Mech	Natr	Anim	OTr	OUn
Croatia	47	(7.0)	(3.4)	(2.3)	(7.3)	(3.4)	(3.3)	(6.0)	(0.6)	1.7	(0.1)	0.5	(0.1)	0.0	0.1	0.3	0.3
Cuba	361	190	(66)	(80)	75	(79)	(37)	(−24)	47	54	13	(6.5)	(2.5)	(0.1)	(4.0)	(1.3)	(0.6)
Cyprus	(23)	(23)	(18)	22	(22)	(19)	(5.0)	5.7	(0.3)	1.0	(0.1)	1.4	(0.2)	(0.0)	0.1	0.2	(0.1)
Czechia	93	(12)	(7.4)	(1.1)	(12)	(1.6)	(2.1)	(1.2)	(1.7)	2.9	0.2	0.9	(0.3)	(0.0)	0.1	(0.8)	0.3
Democratic People's Republic of Korea	346	(289)	(340)	(31)	(43)	(26)	(52)	(8.0)	(39)	16	2.8	19	(5.0)	1.3	1.5	5.6	3.0
Democratic Republic of the Congo	(−83)	(263)	(19)	(196)	371	(76)	(−24)	(−140)	1,057	310	100	129	30	(0.8)	(34)	(20)	(11)
Denmark	(2.1)	(0.9)	(2.8)	(−0.1)	(0.5)	(0.2)	(0.0)	(0.0)	(0.1)	0.3	0.0	(0.1)	(0.0)	(0.0)	(0.0)	(0.0)	0.0
Djibouti	(41)	71	(9.0)	(70)	47	(31)	(23)	(−3.2)	35	(10)	23	(1.5)	(2.6)	(0.1)	(0.0)	(0.7)	(0.0)
Dominica	(0.0)	(0.3)	(0.0)	(0.0)	0.7	(0.2)	(−0.1)	(−0.2)	0.2	0.0	0.1	0.0	(0.0)	(0.1)	(0.0)	(0.0)	(0.0)
Dominican Republic	(106)	75	(9.2)	(−3.7)	82	(38)	(3.6)	(−7.9)	61	14	33	4.3	(1.5)	(0.7)	(0.1)	(0.4)	(0.6)
Ecuador	(3.6)	(17)	(3.0)	(1.6)	38	(31)	(−3.9)	(−2.0)	38	14	11	4.0	(0.7)	(0.6)	(0.5)	(1.0)	(0.6)
Egypt	5,044	(2,374)	(197)	1,952	1,954	(1,816)	(475)	101	(177)	31	39	83	(19)	3.2	3.3	(13)	(17)
El Salvador	(73)	36	(18)	(−12)	102	(126)	(2.0)	(−4.0)	54	17	108	(3.0)	(1.1)	(0.8)	(0.1)	(2.4)	(0.2)
Equatorial Guinea	(−3.4)	(1.6)	(−0.4)	(0.5)	5.2	(0.6)	(−2.1)	(−1.7)	13	4.8	1.8	1.9	2.6	0.0	(0.6)	(0.2)	0.1
Eritrea	(51)	118	(15)	(173)	85	(36)	(29)	(−5.9)	92	(38)	62	(7.7)	(7.4)	(0.0)	(3.2)	(2.0)	(1.0)
Estonia	(1.4)	(0.0)	(0.1)	(0.0)	(0.1)	(0.3)	(1.8)	(0.1)	(0.1)	(0.2)	0.0	0.1	(0.0)	(0.0)	0.0	0.0	0.0
Eswatini	(0.4)	(3.0)	(0.1)	9.4	(4.3)	(1.3)	(−0.6)	0.8	(1.3)	1.8	1.1	1.0	(0.2)	0.0	0.5	0.1	0.2
Ethiopia	(95)	244	(30)	299	187	(182)	(48)	(−16)	210	127	182	(31)	(46)	(0.4)	(15)	(4.9)	(11)
Fiji	(−0.6)	(2.0)	(−0.2)	(0.6)	14	(0.9)	(−0.6)	(−0.2)	0.7	0.9	0.3	0.7	(0.0)	(0.1)	(0.0)	(0.0)	0.1
Finland	(6.3)	(−0.8)	(0.2)	(0.5)	(0.2)	(0.6)	(1.5)	(0.2)	0.3	(1.0)	0.1	0.2	(0.1)	(0.0)	0.0	(0.2)	0.0
France	94	(33)	(20)	(38)	(35)	(28)	(10)	(19)	(3.8)	12	0.4	4.2	(1.2)	0.1	0.4	1.2	2.0
Gabon	(−0.5)	(6.5)	(0.6)	(0.1)	14	(5.4)	(−1.0)	(−2.9)	19	6.6	2.6	2.2	0.4	(0.0)	(0.7)	(0.3)	(0.1)
Gambia	58	78	(8.7)	(44)	45	(33)	(18)	(−9.1)	45	(14)	1.4	(2.4)	(2.9)	(0.0)	(0.2)	(0.7)	(0.1)
Georgia	28	(20)	(1.8)	(4.9)	(5.0)	(1.9)	(9.2)	2.3	(1.7)	1.1	0.2	0.9	(0.4)	0.0	0.1	0.4	0.6
Germany	320	(66)	(83)	(14)	(47)	(84)	(47)	(20)	(7.1)	17	0.8	2.9	(1.2)	(0.0)	0.4	(1.8)	0.7
Ghana	(413)	1,036	(60)	(350)	683	(454)	(179)	(−199)	628	169	62	(23)	(18)	(0.0)	(4.0)	(8.8)	(0.7)
Greece	(99)	(111)	(30)	177	23	(71)	(18)	22	0.9	3.5	(0.4)	8.4	(0.6)	0.1	0.2	0.5	0.4
Greenland	(0.0)	0.0	0.0	(0.0)	(0.0)	(0.0)	(0.0)	(0.0)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Grenada	(0.3)	(0.7)	(0.1)	(−0.4)	2.2	(1.0)	(0.1)	(−0.4)	0.2	0.1	0.1	(0.1)	(0.1)	(0.0)	(0.0)	(0.0)	(0.0)
Guam	(2.4)	(1.0)	(0.5)	(−0.6)	0.9	(1.1)	(0.1)	(−0.1)	0.6	0.6	0.1	(0.0)	(0.0)	(0.0)	(0.0)	(0.0)	(0.0)
Guatemala	(35)	21	(4.5)	(16)	66	(45)	(0.6)	(−1.6)	22	7.2	33	1.3	(3.8)	(0.3)	(0.1)	(0.6)	(1.1)
Guinea	(80)	202	(16)	(211)	128	(57)	(23)	(−27)	135	45	64	(20)	(10)	(0.0)	(4.1)	(2.0)	(2.3)
Guinea-Bissau	(46)	86	(8.9)	(38)	52	(26)	(17)	(−10)	95	(25)	38	(4.1)	(5.2)	(0.0)	(0.1)	(2.5)	(0.1)
Guyana	(3.8)	(8.8)	(0.8)	(−2.0)	15	(5.4)	(2.8)	(−2.9)	4.1	6.7	4.6	(0.8)	(0.2)	(0.0)	(0.3)	(0.2)	(0.1)
Haiti*	(−15)	(60)	(−1.2)	(−8.6)	(78)	(3.7)	(−14)	−18	70	26	62	26	5.3	(1,334)	(0.6)	(2.5)	(3.6)
Honduras*	(35)	(51)	(7.7)	(7.7)	30	(13)	(−0.8)	−3.9	40	9.7	72	5.3	2.2	(117)	(0.5)	(0.7)	(2.0)
Hungary	156	(19)	(18)	(2.4)	(15)	(8.3)	(26)	(13)	(2.4)	5.5	(0.3)	1.2	(0.4)	(0.0)	0.2	(0.7)	0.1
Iceland	(0.0)	0.0	(0.0)	(0.0)	(0.0)	(0.0)	(0.0)	(0.0)	(0.0)	(0.0)	(0.0)	(0.0)	(0.0)	(0.0)	(0.0)	(0.0)	(0.0)
India	60,625	32,726	(35,527)	24,426	29,990	(8,912)	(6,042)	(−1,423)	16,307	9,874	2,846	(1,992)	(843)	(81)	(1,772)	(739)	(277)
Indonesia	(2,740)	6,605	(1,689)	(−650)	2,361	(1,116)	(594)	(−414)	1,690	141	77	(66)	(45)	(156)	(11)	(53)	(6.3)
Iran (Islamic Republic of)	2,558	895	354	637	915	558	650	61	285	62	29	23	42	102	5.0	22	11
Iraq	2,908	1,949	(119)	371	1,237	(719)	(410)	(52)	391	123	265	37	225	(0.2)	(4.3)	(10)	(15)
Ireland	(0.5)	(0.1)	(0.1)	(−0.1)	(0.0)	(0.0)	(0.0)	(0.0)	(0.0)	0.0	(0.0)	(0.0)	(0.0)	(0.0)	(0.0)	(0.0)	(0.0)
Israel	(54)	(64)	(38)	(136)	(91)	(117)	(3.6)	9.4	(2.3)	9.0	(1.2)	2.3	(0.5)	(0.1)	0.2	(0.2)	(0.6)
Italy	223	(102)	(56)	124	109	(72)	(100)	71	5.3	12	0.5	3.1	1.8	0.5	0.7	1.6	0.9
Jamaica	(7.6)	29	(2.9)	(−0.9)	63	(13)	(0.9)	(−4.2)	3.2	1.1	18	0.3	(1.0)	(0.0)	(0.1)	(0.1)	(0.1)
Japan	(436)	(472)	(82)	1,857	86	(375)	(62)	216	9.2	(148)	(2.5)	216	3.1	21	1.4	4.0	3.2
Jordan	(122)	(137)	(27)	87	141	(104)	(55)	3.5	(3.3)	2.6	(4.1)	5.0	(1.8)	(0.2)	0.4	3.6	0.8
Kazakhstan	270	(81)	77	50	(17)	(24)	(15)	103	16	29	7.6	12	3.6	0.1	0.5	1.8	1.5
Kenya	(37)	106	(25)	(82)	96	(61)	(29)	(−9.4)	96	51	65	(7.7)	(5.3)	(0.2)	(4.7)	(2.0)	(1.1)
Kiribati	(0.5)	(1.3)	(0.4)	(−0.3)	3.8	(0.5)	(0.0)	(−0.1)	0.7	0.7	0.1	(0.1)	(0.0)	0.0	(0.0)	(0.0)	(0.0)
Kuwait	399	108	(9.0)	(205)	200	(96)	(99)	(−2.1)	78	(7.5)	4.5	(0.8)	(3.4)	(0.0)	(0.0)	(2.4)	(0.0)
Kyrgyzstan	76	(4.6)	(2.7)	(0.8)	(2.1)	(1.0)	(2.0)	(3.1)	3.2	2.4	0.7	2.1	0.3	0.2	0.1	0.3	0.3
Lao People's Democratic Republic	117	154	(34)	(47)	66	(56)	(19)	(−4.2)	44	13	19	9.7	(2.1)	(0.2)	(0.9)	(1.4)	(1.2)
Latvia	8.0	(1.7)	(0.1)	(−0.1)	(0.2)	(0.0)	(0.3)	(0.5)	(0.3)	(0.5)	0.1	0.3	(0.1)	(0.0)	0.0	0.1	0.1
Lebanon	(13)	(9.0)	(2.4)	21	(13)	(8.6)	(0.6)	1.1	0.3	0.2	(0.3)	0.2	(0.4)	0.0	0.0	0.2	0.6
Lesotho	(0.0)	(0.0)	(−0.1)	(0.0)	(0.1)	(0.0)	(0.0)	(0.0)	(0.1)	(0.1)	(0.1)	0.0	0.0	0.0	0.0	(0.0)	0.0
Liberia	(12)	(38)	(3.2)	(−5.9)	32	(17)	(0.2)	(−7.9)	31	17	4.7	(5.7)	(2.2)	(0.0)	(2.2)	(0.8)	(1.0)
Libya	(228)	(126)	(40)	115	(79)	(145)	(52)	8.6	(22)	6.7	(1.5)	4.7	(3.8)	0.0	0.8	(1.4)	(2.0)

Country	IHD	Str	COPD	LRI	Diab	CKD	HHD	Card	Road	Self	Viol	Drwn	Mech	Natr	Anim	OTr	OUn
Lithuania	20	(1.2)	(0.2)	(−0.3)	(0.2)	(−0.1)	(0.0)	(0.3)	(0.5)	(1.6)	0.2	0.6	(0.1)	(0.0)	0.0	0.1	0.1
Luxembourg	(1.6)	(0.5)	(0.8)	(0.1)	(0.2)	(0.4)	(0.1)	(0.1)	(0.1)	0.1	0.0	0.0	(0.0)	(0.0)	(0.0)	(0.0)	0.0
Madagascar	(31)	111	(9.7)	83	43	(18)	(7.5)	(−4.7)	41	27	5.1	10	3.8	(0.8)	(7.0)	(1.7)	(2.2)
Malawi	(10)	55	(−0.8)	139	43	(18)	(−5.9)	(3.4)	31	34	3.6	16	3.5	0.7	8.1	(1.4)	4.0
Malaysia	(422)	(466)	(148)	(−177)	222	(252)	(14)	(−58)	510	73	24	(15)	(6.4)	(0.1)	(0.7)	(21)	(0.7)
Maldives	(4.1)	(6.5)	(3.6)	(−1.6)	3.7	(4.3)	(0.3)	(−0.8)	0.8	0.4	0.2	(0.2)	(0.1)	(0.2)	(0.0)	(0.0)	(0.0)
Mali	(505)	940	(199)	(987)	758	(362)	(172)	(−86)	724	(165)	408	(52)	(50)	(0.0)	(1.6)	(9.8)	(0.7)
Malta	(3.0)	(2.9)	(1.3)	(7.5)	(2.2)	(2.5)	(0.2)	(1.0)	(0.0)	0.2	(0.0)	(0.1)	(0.0)	(0.0)	0.0	0.0	(0.0)
Marshall Islands	(0.4)	(0.9)	(0.2)	(−0.3)	1.9	(0.4)	(0.0)	(−0.1)	0.6	0.4	0.2	(0.1)	(0.0)	0.0	(0.0)	(0.0)	(0.0)
Mauritania	171	228	(23)	(234)	123	(102)	(71)	(−18)	219	(25)	34	(5.5)	(5.7)	(0.0)	(0.3)	(1.3)	(0.2)
Mauritius	(−0.7)	(1.1)	(−0.9)	(4.4)	(8.3)	(1.6)	(−1.6)	(0.2)	(0.6)	0.8	0.1	0.4	(0.0)	(0.0)	0.0	(0.0)	0.0
Mexico	1,204	405	(249)	573	1,682	933	(61)	(2.3)	266	115	470	26	13	(1.6)	(6.3)	(9.0)	(6.8)
Micronesia (Federated States of)	(1.5)	(2.0)	(0.6)	(−0.6)	2.9	(1.0)	(0.1)	(−0.2)	1.0	0.6	0.3	(0.1)	(0.0)	(0.0)	(0.0)	(0.0)	(0.0)
Monaco	(0.0)	(0.0)	(0.0)	0.0	0.0	(0.0)	(0.0)	0.0	(0.0)	(0.0)	(0.0)	0.0	0.0	0.0	0.0	(0.0)	0.0
Mongolia	(11)	(−3.1)	(0.7)	(1.6)	(0.7)	(1.4)	(0.8)	(1.9)	3.6	(3.9)	(1.2)	(1.0)	(0.5)	(0.1)	(0.1)	(0.1)	(0.2)
Montenegro	3.5	(3.1)	(0.0)	(0.1)	(0.3)	(0.1)	(0.0)	1.0	(0.1)	0.2	0.0	0.1	(0.0)	0.0	0.0	0.0	0.0
Morocco	710	477	(68)	187	153	(208)	(87)	27	11	9.9	(1.6)	8.1	(7.2)	1.1	1.0	2.7	6.2
Mozambique	(4.5)	269	(−1.8)	286	191	(37)	(−22)	(−7.0)	203	185	112	65	11	(4.1)	(27)	(11)	8.8
Myanmar	2,026	3,007	(1,354)	(423)	1,773	(586)	(441)	(−108)	635	(115)	41	(55)	(35)	(243)	(8.1)	(18)	(8.3)
Namibia	(5.9)	(17)	(0.3)	49	18	(3.7)	(−1.2)	(4.0)	(10)	8.1	6.9	3.5	(0.6)	0.3	1.6	(0.3)	0.5
Nauru	(0.3)	(0.3)	(0.1)	(−0.1)	0.4	(0.1)	(0.0)	(0.0)	0.3	0.1	0.1	(0.0)	(0.0)	(0.0)	(0.0)	(0.0)	(0.0)
Nepal	322	260	(204)	225	243	(76)	(28)	(35)	35	102	12	57	(3.9)	(8.9)	8.7	(2.5)	0.9
Netherlands	(14)	(6.7)	(12)	(0.5)	(3.0)	(3.5)	(0.3)	(0.9)	(0.7)	1.3	(0.1)	0.3	(0.1)	(0.0)	(0.0)	(0.1)	0.0
New Zealand	(0.3)	(0.0)	(0.0)	(−0.1)	(0.0)	(0.0)	(0.0)	(0.0)	(0.0)	(0.0)	(0.0)	0.0	(0.0)	(0.0)	(0.0)	(0.0)	0.0
Nicaragua	174	67	(34)	(0.0)	117	(176)	(15)	(−6.3)	46	17	23	(2.5)	(2.0)	(24)	(0.1)	(1.3)	(0.1)
Niger	(808)	1,455	(132)	(2,289)	807	(284)	(164)	(−98)	753	(176)	358	(70)	(71)	(0.0)	(2.2)	(16)	(3.2)
Nigeria	(3,383)	4,012	(493)	(3,475)	3,173	(1,956)	(876)	(−591)	2,779	(1,043)	1,974	(219)	(175)	(0.6)	(22)	(45)	(12)
Niue	(0.0)	(0.0)	(0.0)	(0.0)	0.0	(0.0)	(0.0)	(0.0)	0.0	0.0	0.0	0.0	(0.0)	(0.0)	(0.0)	(0.0)	0.0
North Macedonia	22	(14)	(1.6)	(0.5)	(5.9)	(1.1)	(4.2)	5.5	(0.4)	0.6	0.1	0.2	0.1	0.0	0.0	0.1	0.5
Northern Mariana Islands	(0.7)	(0.9)	(0.5)	(−0.3)	1.3	(0.9)	(0.0)	(−0.1)	0.4	0.2	0.1	(0.0)	(0.0)	(0.0)	(0.0)	(0.0)	(0.0)
Norway	(1.0)	(0.0)	(0.7)	(0.9)	(0.1)	(0.3)	(0.1)	(0.0)	0.1	(0.3)	0.0	0.1	(0.0)	(0.0)	0.0	0.0	0.0
Oman	456	167	(37)	(136)	296	(158)	(165)	(−19)	269	(4.5)	1.6	(2.3)	(7.9)	(0.0)	(0.0)	(1.4)	(0.0)
Pakistan	11,836	6,763	(2,508)	3,044	5,855	(2,662)	(1,300)	(66)	2,487	(803)	510	(491)	(84)	(98)	(208)	(91)	(23)
Palau	(0.3)	(0.4)	(0.2)	(−0.1)	0.8	(0.2)	(0.0)	(0.0)	0.2	0.1	0.1	(0.0)	(0.0)	0.0	(0.0)	(0.0)	(0.0)
Palestine	(79)	(76)	(11)	46	(70)	(51)	(13)	7.6	(1.7)	0.9	(2.9)	1.7	(1.5)	0.0	0.1	0.1	(0.4)
Panama	(8.2)	(23)	(9.5)	(−9.1)	43	(28)	(0.4)	(−8.1)	26	8.3	22	(2.3)	(0.6)	(0.1)	(0.3)	(0.3)	(0.1)
Papua New Guinea	(13)	46	(24)	(−12)	70	(4.7)	(−0.6)	−5.0	32	3.9	15	(9.7)	(1.5)	(3.1)	(0.2)	(1.0)	(1.5)
Paraguay	(65)	(67)	(3.9)	127	(124)	(34)	(3.3)	12	(25)	13	16	7.0	(1.0)	0.2	1.0	(0.6)	(1.7)
Peru	(14)	23	(3.8)	90	32	(46)	(0.0)	(−0.8)	30	4.9	4.8	3.2	1.5	(0.5)	(0.2)	(1.3)	(1.2)
Philippines	(2,222)	2,159	(556)	(−324)	1,703	(1,432)	(625)	(−225)	865	193	764	(79)	(24)	(16)	(2.8)	(29)	(1.7)
Poland	256	(63)	(18)	(11)	(26)	(5.4)	(14)	(25)	(8.8)	12	1.1	4.4	(1.3)	(0.0)	0.3	(2.6)	0.8
Portugal	(4.6)	(6.0)	(1.1)	20	5.8	(3.4)	(0.8)	2.0	(0.2)	1.3	(0.1)	0.3	0.0	0.0	0.1	0.2	0.1
Puerto Rico	(11)	8.2	(7.9)	(0.6)	42	(25)	(−0.7)	(−2.1)	5.5	3.5	5.3	0.3	(0.2)	(2.9)	(0.1)	(0.1)	(0.0)
Qatar	252	110	(28)	(68)	379	(131)	(46)	(−9.6)	173	(15)	2.7	(1.4)	(9.6)	0.0	(0.0)	(3.0)	(0.0)
Republic of Korea	198	(134)	(112)	(345)	151	(136)	(72)	33	(23)	(88)	(1.7)	34	2.4	1.1	1.2	(8.8)	1.7
Republic of Moldova	105	(16)	(3.5)	(0.8)	(3.2)	(0.6)	(8.3)	(1.8)	(1.1)	1.9	(0.6)	1.6	(0.2)	(0.0)	0.1	(0.3)	0.5
Romania	252	(94)	(18)	(15)	(13)	(8.9)	(57)	(73)	(5.8)	6.4	0.8	6.4	1.3	0.1	0.6	2.4	0.6
Russian Federation	1,487	(197)	(95)	(87)	(128)	(57)	(63)	(153)	78	(154)	66	39	18	0.7	2.7	17	22
Rwanda	(−0.1)	(0.0)	(−0.1)	0.3	(0.0)	(0.0)	−0.1	(0.0)	0.1	0.1	0.0	0.1	(0.0)	0.0	0.0	(0.0)	0.0
Saint Kitts and Nevis	(0.2)	(0.6)	(0.1)	(−0.1)	0.8	(0.5)	(0.0)	(−0.2)	0.1	0.0	0.1	(0.0)	(0.0)	(0.0)	(0.0)	(0.0)	(0.0)
Saint Lucia	(0.6)	(2.4)	(0.7)	(−0.7)	4.7	(1.9)	(0.5)	(−1.1)	0.9	0.4	0.7	(0.1)	(0.1)	(0.0)	(0.0)	(0.0)	(0.0)
Saint Vincent and the Grenadines	(0.4)	(0.9)	(0.1)	(−0.4)	2.2	(0.6)	(0.2)	(−0.2)	0.2	0.2	0.5	(0.1)	(0.0)	(0.0)	(0.0)	(0.0)	(0.0)
Samoa	(0.0)	(0.9)	(0.2)	(−0.2)	1.6	(0.4)	(−0.1)	(−0.1)	0.7	0.6	0.2	0.3	(0.0)	(0.3)	(0.0)	(0.0)	(0.0)
San Marino	(0.1)	(0.0)	(0.0)	0.1	0.0	(0.0)	(0.0)	0.1	(0.0)	(0.0)	(0.0)	0.0	0.0	0.0	0.0	(0.0)	0.0
Sao Tome and Principe	(0.3)	(0.8)	(0.2)	(0.3)	0.3	(0.8)	(0.0)	(−0.1)	0.4	0.1	0.1	0.0	(0.0)	0.0	(0.0)	(0.0)	(0.0)
Saudi Arabia	3,710	1,748	(421)	1,292	1,404	(2,307)	(1,320)	(−53)	2,430	(223)	285	(36)	(90)	(0.4)	(2.4)	(22)	(11)
Senegal	358	503	(71)	(202)	372	(199)	(106)	(−67)	295	(133)	16	(20)	(18)	(0.1)	(2.5)	(4.0)	(1.5)
Serbia	109	(43)	(7.3)	(1.8)	(16)	(6.7)	(13)	(17)	(1.6)	4.8	0.4	1.0	(0.3)	0.0	0.1	0.8	0.3
Seychelles	(0.4)	(0.8)	(0.3)	(−0.1)	0.7	(0.8)	(0.5)	(−0.2)	0.4	0.2	0.1	(0.2)	(0.0)	(0.0)	(0.0)	(0.0)	(0.0)

Country	IHD	Str	COPD	LRI	Diab	CKD	HHD	Card	Road	Self	Viol	Drwn	Mech	Natr	Anim	OTr	OUn
Sierra Leone	(79)	132	(13)	(50)	75	(36)	(16)	(−21)	105	29	6.6	(10)	(6.1)	(1.1)	(1.9)	(4.9)	(1.0)
Singapore	(−31)	(10)	(4.5)	(175)	(5.0)	(11)	(1.9)	(−8.3)	8.1	16	1.4	(1.0)	(0.7)	0.0	(0.0)	(0.1)	(0.0)
Slovakia	75	(9.4)	(1.9)	(0.6)	(2.6)	(1.2)	(1.7)	(1.5)	(1.1)	1.7	0.2	0.9	(0.3)	0.0	0.1	0.5	0.1
Slovenia	8.1	(2.7)	(1.8)	(0.7)	(1.4)	(0.8)	(3.0)	(3.3)	(0.3)	0.8	0.0	0.1	(0.1)	(0.0)	0.0	(0.1)	0.0
Solomon Islands	(0.2)	(5.2)	(0.7)	(−2.4)	9.8	(0.7)	(−0.3)	−0.4	4.4	3.7	1.2	(1.9)	(0.2)	(0.1)	(0.0)	(0.1)	(0.2)
Somalia	(43)	(438)	(59)	(611)	413	(109)	(48)	(−56)	942	195	641	(72)	(110)	(12)	(24)	(26)	(11)
South Africa	(31)	(155)	(29)	363	(196)	(58)	(−3.5)	64	(24)	38	(55)	20	(5.2)	0.3	8.2	2.4	2.9
South Sudan	(145)	327	(42)	(441)	305	(177)	(56)	(−27)	276	(112)	352	(28)	(28)	(0.1)	(4.0)	(8.3)	(1.3)
Spain	149	127	(142)	242	86	120	(38)	101	2.1	17	(0.7)	6.5	0.8	0.2	0.5	1.3	0.9
Sri Lanka	402	650	(339)	(−11)	618	(180)	(70)	(−39)	125	122	18	(12)	(7.0)	(45)	(1.5)	(5.4)	(0.4)
Sudan	5,240	2,224	(333)	(618)	738	(875)	(1,074)	(−50)	1,370	(211)	171	(29)	(146)	(0.3)	(0.1)	(12)	(2.5)
Suriname	(8.6)	(12)	(2.1)	(−2.8)	14	(9.5)	(2.0)	(−1.6)	6.5	6.0	2.2	(0.4)	(0.3)	0.0	(0.0)	(0.1)	(0.0)
Sweden	6.1	(0.6)	(0.8)	(0.0)	(0.5)	(0.5)	(0.2)	(0.1)	(0.2)	(0.9)	0.1	0.1	(0.1)	(0.0)	0.0	0.1	0.0
Switzerland	(22)	(4.5)	(4.9)	(0.4)	(2.4)	(4.5)	(2.7)	(0.7)	(0.5)	1.6	0.1	0.3	(0.2)	(0.1)	(0.0)	(0.3)	0.3
Syrian Arab Republic	1,034	379	90	183	127	261	(104)	22	(6.2)	2.1	(7.2)	4.9	(2.1)	0.1	0.7	2.4	1.9
Taiwan	(92)	(167)	(48)	917	268	(210)	(4.9)	20	41	72	1.9	24	(1.7)	2.4	0.3	(1.2)	3.5
Tajikistan	218	82	16	51	22	2.4	20	3.2	2.0	3.6	0.9	13	0.2	3.5	0.0	0.8	0.4
Thailand	(2,166)	3,669	(1,311)	(1,740)	2,396	(2,553)	(211)	(−82)	1,831	(451)	203	(101)	(30)	(5.4)	(1.5)	(32)	(1.0)
Timor-Leste	(1.4)	4.8	(0.5)	(0.3)	1.6	(1.2)	(−0.3)	(−0.4)	1.9	0.9	0.5	0.9	(0.1)	(0.0)	(0.1)	(0.1)	0.1
Togo	(109)	214	(23)	(92)	125	(72)	(44)	(−32)	161	63	59	(8.3)	(8.6)	(0.0)	(0.7)	(3.1)	(0.3)
Tokelau	(0.0)	(0.0)	(0.0)	(0.0)	0.0	(0.0)	(0.0)	(0.0)	0.0	0.0	0.0	(0.0)	(0.0)	0.0	(0.0)	(0.0)	(0.0)
Tonga	(0.1)	(0.3)	(0.0)	(0.1)	1.2	(0.1)	(0.0)	(0.0)	0.2	0.1	0.0	0.1	(0.0)	(0.0)	(0.0)	(0.0)	0.0
Trinidad and Tobago	(6.2)	(12)	(1.8)	(−4.0)	45	(10)	(0.3)	(−2.5)	5.8	4.4	6.3	(0.6)	(0.2)	(0.0)	(0.1)	(0.1)	(0.0)
Tunisia	353	254	(60)	140	99	156	(92)	22	(7.5)	5.3	(0.6)	3.9	(6.9)	0.1	0.6	0.6	1.7
Türkiye	700	339	331	389	363	258	216	31	17	17	5.5	9.2	6.2	8.6	0.9	5.5	3.7
Turkmenistan	378	163	14	56	47	31	28	66	1.3	12	(2.9)	8.4	(1.3)	0.0	0.4	1.2	2.1
Tuvalu	(0.3)	(0.3)	(0.1)	(−0.1)	0.4	(0.1)	(0.0)	(0.0)	0.2	0.1	0.0	(0.0)	(0.0)	0.0	(0.0)	(0.0)	(0.0)
Uganda	(−11)	(23)	(−4.5)	53	39	(7.1)	(−10)	(−4.5)	64	40	26	19	4.4	(0.3)	10	(2.8)	3.6
Ukraine	2,063	(217)	(32)	(3.1)	(9.4)	(3.0)	(14)	(68)	(24)	45	15	28	(4.2)	0.1	0.8	5.8	4.9
United Arab Emirates	1,409	708	(233)	(765)	769	(490)	(464)	(−71)	542	(37)	8.9	(3.9)	(25)	0.0	(0.0)	(7.4)	(0.0)
United Kingdom	(25)	(5.7)	(10)	(0.1)	(1.3)	(−0.3)	(−0.5)	(0.4)	(0.5)	1.3	(0.1)	0.2	(0.1)	(0.0)	(0.0)	(0.1)	0.0
United Republic of Tanzania	(37)	107	(4.2)	(95)	120	(57)	(−6.5)	(−11)	91	63	49	25	4.9	(0.4)	(11)	(2.8)	2.6
United States of America	4,197	1,749	2,195	1,951	1,085	2,420	636	898	220	473	106	97	29	7.8	9.4	35	7.4
United States Virgin Islands	(1.7)	(0.7)	(0.2)	(−0.3)	1.4	(0.7)	(0.1)	(−0.3)	0.3	0.1	0.4	(0.1)	(0.0)	(0.0)	(0.0)	(0.0)	(0.0)
Uruguay	(1.7)	(3.3)	(0.2)	6.8	(2.1)	(1.1)	(0.0)	1.6	(0.1)	1.3	(0.2)	0.3	(0.1)	0.0	0.0	0.1	0.1
Uzbekistan	2,039	384	47	205	271	112	100	41	16	46	(5.5)	47	2.3	0.2	1.1	2.5	7.9
Vanuatu	(0.5)	1.4	(0.1)	(0.1)	2.3	(0.4)	(−0.1)	(−0.1)	1.1	1.1	0.0	0.7	0.0	(0.2)	(0.0)	(0.0)	0.1
Venezuela (Bolivarian Republic of)	1,521	529	(147)	(28)	857	(508)	(131)	(−38)	486	75	376	(10)	(39)	(84)	(1.1)	(12)	(0.5)
Viet Nam	2,027	7,493	(1,012)	904	2,603	(1,393)	(531)	(−94)	1,775	575	104	337	(134)	(10.0)	(43)	(45)	(37)
Yemen	1,102	656	(76)	179	136	(110)	(209)	(−9.3)	538	(51)	33	(12)	(36)	(0.5)	(0.8)	(6.3)	(5.4)
Zambia	(14)	(65)	(−0.3)	149	44	(29)	(−8.0)	13	27	34	27	18	(1.9)	0.1	8.8	(3.0)	2.2
Zimbabwe	(13)	(59)	(2.1)	235	47	(18)	(−4.9)	15	15	34	13	18	(2.4)	0.8	8.8	1.5	4.2

Table 13. Cold deaths per year by country and cause, on the same basis and with the same abbreviations as Table 12.

Country	IHD	Str	COPD	LRI	Diab	CKD	HHD	Card	Road	Self	Viol	Drwn	Mech	Natr	Anim	OTr	OUn
Afghanistan	3,454	1,827	729	1,544	553	1,114	1,256	67	−363	−127	−312	−280	−224	−126	−26	−42	−325
Albania	583	703	163	122	22	73	88	37	−3.9	−6.4	−3.5	−5.6	−4.5	−0.3	−0.5	−3.1	−3.1
Algeria	6,993	2,716	1,448	1,045	813	2,206	3,232	76	−103	−85	−12	−69	−43	−27	−9.1	−29	−40
American Samoa	0.1	(0.0)	(0.0)	(0.0)	(0.0)	(0.1)	(0.0)	(0.0)	0.0	0.0	0.0	(0.0)	(0.0)	(0.0)	(0.0)	(0.0)	(0.0)
Andorra	5.9	5.5	9.5	6.4	1.9	5.1	3.6	1.4	−0.9	−0.6	0.0	0.0	0.0	0.0	0.0	−0.3	0.0
Angola	417	269	127	211	117	201	367	(55)	−124	−129	−27	−147	−6.7	−1.8	−31	−9.8	−12
Antigua and Barbuda	0.3	(0.0)	(0.0)	(0.1)	(−0.1)	(0.2)	(0.3)	(0.1)	0.0	0.0	0.0	(0.0)	(0.0)	(0.0)	(0.0)	(0.0)	(0.0)
Argentina	2,531	1,442	2,495	6,095	737	2,200	1,371	436	−52	−267	−65	−66	−23	−2.1	−7.6	−22	−50
Armenia	678	225	165	257	52	63	113	19	−17	−8.4	−3.2	−6.3	−3.4	0.0	−0.9	−1.1	−3.8

Country	IHD	Str	COPD	LRI	Diab	CKD	HHD	Card	Road	Self	Viol	Drwn	Mech	Natr	Anim	OTr	OUn
Australia	2,002	991	1,859	993	442	1,088	261	88	−25	−211	−9.1	−34	−6.4	−1.6	−6.0	−13	−6.0
Austria	1,045	410	659	413	203	763	541	200	−26	−68	−2.6	−8.9	−7.2	−1.4	−2.0	−6.8	−1.1
Azerbaijan	3,680	1,227	439	338	222	213	489	159	−15	−20	−15	−43	−3.5	−1.4	−1.0	−2.6	−6.1
Bahamas	6.3	(1.4)	(0.8)	(0.8)	(0.8)	3.8	(7.0)	(1.1)	−1.3	−0.3	−1.9	−0.6	(0.0)	(−0.1)	(−0.1)	(−0.1)	0.0
Bahrain	79	(12)	(16)	(−2.9)	(47)	(40)	(12)	(1.7)	−10	−2.1	−0.9	(−0.3)	(−0.3)	0.0	(0.0)	(−0.4)	(−0.1)
Bangladesh	5,333	2,789	(1,772)	(27)	(1,024)	893	(845)	(314)	−275	−168	−88	−137	(−4.4)	(−35)	(−37)	(−32)	(−6.4)
Barbados	0.5	(0.1)	(0.0)	(0.2)	(−0.2)	(0.3)	(0.2)	(0.1)	−0.1	0.0	−0.1	(0.0)	(0.0)	(0.0)	(0.0)	(0.0)	(0.0)
Belarus	3,336	1,187	233	157	52	43	27	85	−55	−109	−22	−56	−14	−0.1	−2.1	−2.8	−13
Belgium	633	636	1,102	1,887	133	660	130	120	−28	−82	−4.8	−13	−4.8	−0.2	−1.7	−6.1	−2.9
Belize	4.8	(1.2)	(0.9)	(0.0)	(1.4)	4.1	(1.5)	(0.3)	−1.5	−0.5	−1.5	−0.3	(−0.1)	(0.0)	(−0.1)	(−0.1)	(0.0)
Benin	25	(15)	(2.3)	(5.5)	(1.7)	(24)	(5.8)	(2.7)	−19	−5.8	−7.8	(−1.1)	(−1.1)	(0.0)	(0.0)	(−0.3)	(0.0)
Bermuda	5.0	1.0	1.1	(0.5)	(0.7)	1.4	1.2	(0.3)	(−0.1)	−0.1	0.0	−0.2	(0.0)	0.0	0.0	(0.0)	0.0
Bhutan	50	32	80	17	16	19	16	3.0	−1.7	−3.0	−0.3	−3.1	−3.0	−1.0	−1.5	−0.4	−0.5
Bolivia (Plurinational State of)	444	402	331	904	274	754	162	12	−72	−34	−20	−44	−11	−4.4	−5.7	−23	−22
Bosnia and Herzegovina	596	514	203	106	179	103	157	124	−5.0	−14	−0.7	−1.3	−0.5	−0.1	−0.1	−2.2	−0.3
Botswana	63	45	41	43	44	35	56	13	−7.2	−15	−6.4	−7.6	(−1.0)	−0.1	−4.5	−0.4	−1.2
Brazil	5,549	2,538	3,634	2,941	1,761	2,353	1,995	530	−422	−543	−983	−372	−35	−13	−64	−30	−98
Brunei Darussalam	0.3	(0.0)	(0.1)	(0.1)	(−0.1)	(0.2)	(0.1)	(0.1)	−0.1	0.0	0.0	(0.0)	(0.0)	(0.0)	(0.0)	(0.0)	(0.0)
Bulgaria	1,319	1,450	285	292	139	251	1,811	58	−18	−36	−4.2	−18	−6.8	−0.2	−2.3	−8.2	−5.8
Burkina Faso	71	(24)	(4.9)	(18)	(7.8)	(43)	(22)	(6.8)	−52	−13	−24	(−1.8)	(−1.6)	(0.0)	(0.4)	(−0.7)	(0.0)
Burundi	284	235	121	171	81	113	116	(6.9)	−20	−63	−2.3	−65	−9.3	−1.1	−42	−5.5	−25
Cabo Verde	14	5.1	2.1	(2.9)	(1.6)	3.8	3.5	(0.4)	−0.7	−3.2	−1.5	−2.9	(−0.1)	0.0	−0.5	(0.0)	−0.2
Cambodia	53	(29)	(10)	(−3.3)	(4.4)	(15)	(12)	(1.1)	−16	−3.0	−2.0	(−1.3)	(−0.7)	(0.0)	(0.0)	(−0.4)	(0.0)
Cameroon	275	181	62	(109)	73	212	119	(22)	−100	−74	−15	−53	−5.8	−0.3	−22	−5.7	−8.3
Canada	3,347	1,992	3,744	3,226	739	2,062	470	195	−181	−358	−35	−47	−24	−2.0	−4.7	−53	−7.2
Central African Republic	24	(7.8)	(3.7)	(6.8)	(3.1)	(6.6)	(8.3)	(2.8)	−20	−5.2	−3.7	(−3.3)	−0.7	(0.0)	(−0.7)	(−0.3)	(−0.3)
Chad	77	(42)	(7.5)	(37)	(9.5)	(28)	(17)	(6.8)	−48	−14	−25	(−5.6)	(−3.5)	(0.0)	(−0.1)	(−0.7)	(−0.4)
Chile	862	1,081	1,462	1,884	378	1,368	923	63	−67	−131	−21	−48	−28	−7.7	−5.2	−13	−12
China	165,841	219,943	268,422	43,755	16,886	31,241	73,522	3,680	−5,425	−10,723	−456	−10,332	−1,524	−489	−538	−700	−626
Colombia	2,807	822	2,653	777	345	934	570	(30)	−82	−118	−244	−76	−12	−18	−23	−17	−12
Comoros	2.8	(1.2)	(0.7)	(1.3)	(0.2)	1.8	1.9	0.3	−1.8	−0.7	−0.8	−0.5	(−0.1)	(0.0)	(−0.3)	(0.0)	−0.1
Congo	39	(11)	(7.0)	(8.8)	(3.4)	16	22	(5.5)	−22	−10	−3.8	−4.4	(−0.6)	(0.0)	(−1.4)	(−0.6)	−0.3
Cook Islands	0.3	(0.1)	(0.1)	(0.0)	(0.2)	(0.1)	0.4	(0.0)	−0.1	−0.1	0.0	0.0	(0.0)	(0.0)	0.0	(0.0)	0.0
Costa Rica	151	48	131	36	37	111	60	(3.6)	−6.2	−18	−10	−12	−0.5	−1.0	−1.2	−2.0	−1.3
Côte d'Ivoire	89	(28)	(6.4)	(19)	(3.5)	(43)	(21)	(9.6)	−41	−19	−24	(−3.3)	(−2.4)	(0.0)	(−0.7)	(−0.7)	(−0.3)
Croatia	845	419	359	195	124	198	219	73	−7.6	−25	−1.3	−7.3	−2.3	0.0	−0.9	−4.3	−3.7
Cuba	366	(46)	(45)	(22)	(5.5)	55	(50)	(19)	−26	−34	−7.9	−4.7	(−1.4)	(−0.1)	(−3.2)	(−0.8)	(−0.5)
Cyprus	115	43	79	24	42	57	39	6.3	(−0.8)	−3.5	−0.4	−4.9	(−0.4)	0.0	−0.2	−0.8	−0.2
Czechia	1,916	704	673	1,036	453	316	393	67	−34	−58	−4.0	−16	−5.8	−0.3	−1.8	−15	−5.5
Democratic People's Republic of Korea	3,610	5,914	7,599	1,773	433	658	1,454	96	−387	−173	−30	−165	−50	−11	−13	−52	−26
Democratic Republic of the Congo	683	409	337	484	189	315	614	99	−288	−223	−50	−199	−18	−1.6	−64	−21	−22
Denmark	252	323	900	804	135	373	57	21	−14	−34	−1.8	−5.3	−2.7	0.0	−0.6	−3.4	−3.6
Djibouti	15	(7.3)	(1.9)	(−2.4)	(2.8)	(10)	(5.3)	(0.4)	−5.5	−1.6	−3.5	(−0.2)	(−0.4)	(0.0)	(0.0)	(−0.1)	(0.0)
Dominica	0.2	(0.0)	(0.0)	(0.1)	(−0.1)	(0.1)	(0.2)	0.1	−0.1	0.0	0.0	0.0	(0.0)	(0.0)	(0.0)	(0.0)	(0.0)
Dominican Republic	230	39	19	(4.7)	(9.1)	40	37	(4.3)	−27	−11	−19	−6.3	−0.9	−1.0	−0.1	−0.3	−1.2
Ecuador	663	392	415	880	300	858	271	13	−91	−55	−35	−41	−5.7	−7.3	−6.7	−16	−9.8
Egypt	15,981	3,440	1,100	(671)	2,365	5,047	3,108	106	−318	−80	−88	−229	(−36)	−9.2	−9.8	−31	−49
El Salvador	33	(2.5)	(4.1)	(0.8)	(1.3)	29	(1.5)	(0.4)	−6.1	−2.5	−13	−0.8	−0.1	(−0.2)	(0.0)	(−0.3)	(−0.1)
Equatorial Guinea	5.3	(1.4)	1.1	(2.7)	(0.6)	2.5	3.6	1.0	−4.1	−2.4	−0.8	−1.2	(−0.9)	0.0	−0.4	(−0.1)	−0.1
Eritrea	79	74	35	79	32	47	53	(4.1)	−17	−24	−22	−19	−3.2	0.0	−15	−1.8	−5.3
Estonia	129	70	32	41	17	81	306	16	−5.1	−13	−3.0	−3.8	−1.2	0.0	−0.2	−1.8	−0.7
Eswatini	28	26	20	31	26	24	28	2.8	−4.0	−11	−5.3	−6.6	−0.9	0.0	−3.3	−0.7	−1.1
Ethiopia	1,380	1,825	906	2,345	740	2,163	1,230	78	−127	−520	−372	−481	−164	−9.2	−353	−44	−266
Fiji	22	(3.7)	3.8	(−0.1)	(11)	(4.5)	3.3	(0.3)	−1.0	−1.7	−0.5	−1.8	(0.0)	−0.2	−0.1	(−0.1)	−0.2
Finland	662	428	273	226	44	158	352	42	−21	−64	−5.6	−15	−3.7	−0.1	−1.3	−11	−3.0
France	3,794	3,622	3,618	6,978	1,459	3,488	1,846	670	−119	−390	−13	−137	−42	−3.8	−12	−44	−66
Gabon	6.9	(2.0)	(1.5)	(2.5)	(0.7)	4.9	5.6	1.3	−5.6	−2.6	−0.9	−1.2	−0.1	(0.0)	(−0.4)	(−0.1)	−0.1
Gambia	9.1	(3.1)	(0.8)	(1.2)	(0.5)	(4.4)	(2.4)	(0.9)	−3.6	−1.4	−0.1	(−0.4)	(−0.2)	(0.0)	(−0.1)	(−0.1)	(−0.1)
Georgia	458	719	117	297	80	93	413	29	−23	−16	−2.8	−12	−6.3	−0.4	−1.2	−6.3	−7.6

Country	IHD	Str	COPD	LRI	Diab	CKD	HHD	Card	Road	Self	Viol	Drwn	Mech	Natr	Anim	OTr	OUn
Germany	8,183	4,255	6,984	6,009	1,819	7,883	5,684	1,061	−178	−452	−20	−68	−29	−1.0	−10	−45	−16
Ghana	93	(45)	(5.0)	(37)	(−4.7)	(75)	(33)	(25)	−67	−19	−6.9	(−2.5)	(−2.0)	(0.0)	(−0.3)	(−1.0)	(−0.2)
Greece	1,155	846	654	871	121	817	438	46	−7.8	−24	−2.0	−58	−3.8	−0.8	−1.7	−4.0	−2.5
Greenland	5.5	7.9	9.7	6.2	(0.6)	(2.3)	0.8	(0.5)	−0.3	−3.9	−0.5	−0.6	−0.1	0.0	(0.0)	−0.6	0.0
Grenada	0.1	(0.0)	(0.0)	(0.1)	(−0.1)	(0.1)	(0.1)	(0.0)	0.0	0.0	0.0	(0.0)	(0.0)	(0.0)	(0.0)	(0.0)	(0.0)
Guam	0.3	(0.0)	(0.0)	(0.0)	(0.0)	(0.1)	(0.0)	(0.0)	0.0	0.0	0.0	(0.0)	(0.0)	(0.0)	(0.0)	(0.0)	(0.0)
Guatemala	628	167	146	392	248	498	44	(4.2)	−18	−34	−79	−21	−15	−5.3	−2.1	−5.8	−27
Guinea	61	(25)	(6.2)	(33)	(2.9)	(24)	(15)	(8.5)	−29	−12	−15	(−7.0)	(−2.3)	(0.0)	(−1.9)	(−0.5)	(−1.2)
Guinea-Bissau	2.8	(1.2)	(0.2)	(0.1)	(0.3)	(1.1)	(0.7)	(0.2)	−1.8	−0.5	−0.7	(−0.1)	(−0.1)	(0.0)	(0.0)	(0.0)	(0.0)
Guyana	1.1	(0.3)	(0.1)	(0.2)	(−0.1)	(0.6)	(0.8)	(0.2)	−0.3	−0.5	−0.3	(−0.1)	(0.0)	(0.0)	(0.0)	(0.0)	(0.0)
Haiti*	190	(50)	22	(4.5)	(7.2)	(26)	36	(10.0)	−34	−19	−36	−30	(−3.0)	(201)	−0.8	(−1.9)	−4.6
Honduras*	268	79	90	(2.2)	(14)	45	62	(3.7)	−26	−14	−69	−14	−2.3	(−0.2)	−1.5	−1.3	−6.5
Hungary	2,054	740	926	334	207	345	1,047	190	−24	−68	−3.7	−14	−4.9	−0.2	−2.9	−8.5	−0.6
Iceland	31	22	31	38	(3.1)	(11)	5.5	(1.1)	−1.6	(−5.4)	−0.2	−0.6	−0.3	(−0.8)	(0.0)	−1.3	−0.1
India	59,002	11,670	33,414	(2,052)	7,647	8,513	(6,026)	(2,170)	−7,195	−5,571	−1,384	−1,676	−399	(−83)	(−1,870)	(−432)	(−306)
Indonesia	1,716	1,433	1,129	(−98)	(−60)	(325)	855	(40)	−253	−57	−19	−100	−9.9	−57	−46	−21	−23
Iran (Islamic Republic of)	12,378	4,475	3,710	2,698	2,644	3,632	5,555	167	−598	−223	−85	−116	−175	−428	−27	−126	−60
Iraq	4,520	1,475	239	131	833	1,285	800	124	−262	−129	−215	−58	−182	−0.4	−8.3	−12	−30
Ireland	330	208	542	772	63	215	52	59	−8.1	−21	−1.1	−8.0	−2.3	−0.1	−0.7	−2.2	−0.3
Israel	341	124	190	135	184	379	44	12	−5.3	−34	−4.1	−8.1	(−1.0)	−0.3	−0.9	−0.7	−2.2
Italy	6,036	4,368	5,662	2,854	1,675	3,299	5,690	450	−93	−198	−8.4	−52	−34	−8.4	−11	−32	−16
Jamaica	11	(4.5)	(1.7)	(0.8)	(1.7)	(6.4)	(7.2)	(1.4)	−0.9	−0.3	−4.9	−0.1	(−0.3)	(0.0)	(0.0)	(0.0)	(0.0)
Japan	7,100	8,874	5,114	16,835	668	8,178	2,394	695	−80	−1,073	−18	−1,789	−30	−168	−11	−40	−27
Jordan	872	378	203	163	376	488	646	4.9	(−10)	−11	−13	−20	−5.2	−0.7	−1.6	−15	−3.2
Kazakhstan	3,035	2,653	1,773	924	132	348	239	653	−190	−301	−91	−109	−40	−1.1	−4.4	−18	−14
Kenya	637	789	628	1,087	337	690	740	(23)	−58	−226	−140	−153	−19	−4.9	−160	−18	−46
Kiribati	0.3	(0.0)	(0.0)	(0.0)	(0.0)	(0.1)	(0.0)	(0.0)	0.0	0.0	0.0	(0.0)	(0.0)	0.0	(0.0)	(0.0)	(0.0)
Kuwait	340	(29)	(5.7)	(−14)	(39)	(83)	(58)	(1.8)	−38	−4.0	−2.2	(−0.5)	(−1.6)	(0.0)	(0.0)	(−1.2)	(0.0)
Kyrgyzstan	1,534	595	420	150	45	98	190	51	−92	−56	−19	−35	−7.1	−4.1	−1.5	−5.5	−5.1
Lao People's Democratic Republic	224	111	77	(19)	27	82	58	5.4	−22	−13	−13	−19	−1.5	−0.4	−2.3	−1.5	−3.1
Latvia	328	318	44	60	32	44	120	62	−9.6	−18	−4.8	−10	−2.5	0.0	−0.4	−3.2	−1.7
Lebanon	516	213	266	319	193	339	138	6.4	−6.6	−2.7	−3.6	−3.4	−6.1	−0.2	−0.3	−3.3	−9.2
Lesotho	61	157	105	309	87	72	101	11	−30	−22	−27	−17	−5.5	−0.2	−9.0	−4.0	−4.8
Liberia	14	(3.5)	(1.1)	(4.6)	(−0.7)	(6.0)	(4.6)	(2.0)	−6.5	−3.7	−1.0	(−1.2)	(−0.5)	(0.0)	(−0.5)	(−0.2)	(−0.2)
Libya	880	196	157	72	125	368	334	8.3	−47	−23	−4.8	−15	(−7.3)	−0.1	−2.7	−4.1	−6.5
Lithuania	650	297	96	85	34	46	105	39	−13	−36	−5.2	−14	−2.4	0.0	−0.8	−2.9	−2.2
Luxembourg	45	35	64	56	10	44	16	5.8	−2.0	−3.6	−0.2	−0.5	−0.4	0.0	−0.2	−0.3	−0.5
Madagascar	661	904	462	670	172	253	723	43	−49	−129	−15	−109	−17	−10	−88	−15	−30
Malawi	271	242	100	187	105	174	151	17	−47	−110	−8.8	−67	−9.1	−2.9	−36	−5.4	−18
Malaysia	65	(6.8)	(10)	(55)	(−7.5)	(26)	(4.0)	(4.4)	−39	−7.8	−2.2	(−3.2)	(−0.6)	(−0.1)	(−0.4)	(−1.9)	(−0.4)
Maldives	(0.3)	(0.0)	(0.1)	(−0.1)	(−0.1)	(0.2)	(0.0)	(0.0)	0.0	0.0	0.0	(0.0)	(0.0)	(0.0)	(0.0)	(0.0)	(0.0)
Mali	98	(50)	(23)	(−13)	(25)	(60)	(21)	(5.6)	−51	−12	−28	(−3.3)	(−3.5)	(0.0)	(0.1)	(−0.7)	(0.0)
Malta	46	11	14	19	8.4	17	8.7	(2.1)	(−0.1)	−1.2	−0.1	−0.5	(−0.1)	0.0	0.0	−0.1	−0.1
Marshall Islands	0.0	(0.0)	(0.0)	(0.0)	(0.0)	(0.0)	(0.0)	(0.0)	0.0	0.0	0.0	(0.0)	(0.0)	0.0	(0.0)	(0.0)	(0.0)
Mauritania	40	(16)	(3.6)	(−3.5)	(5.8)	(23)	(11)	(2.2)	−22	−2.9	−3.5	(−1.0)	(−0.6)	(0.0)	(−0.2)	(−0.2)	(−0.1)
Mauritius	34	(8.6)	12	(0.9)	(19)	38	17	(1.7)	(−3.0)	−4.6	−0.3	−2.4	(0.0)	(0.0)	0.0	(−0.2)	0.0
Mexico	10,624	2,330	4,848	3,104	5,593	8,263	1,441	69	−299	−458	−1,131	−251	−54	−18	−74	−79	−91
Micronesia (Federated States of)	0.0	(0.0)	(0.0)	(0.0)	(0.0)	(0.0)	(0.0)	(0.0)	0.0	0.0	0.0	(0.0)	(0.0)	(0.0)	(0.0)	(0.0)	(0.0)
Monaco	3.4	3.3	2.7	4.3	0.3	2.0	1.5	0.5	(0.0)	−0.3	0.0	−0.1	0.0	0.0	0.0	0.0	0.0
Mongolia	446	443	116	89	(13)	(47)	32	(27)	−85	−87	−33	−22	−12	(−1.8)	(−1.7)	−3.4	−3.8
Montenegro	103	182	9.5	14	14	25	18	24	−2.2	−4.5	−0.8	−1.2	−0.2	0.0	0.0	−0.5	−0.4
Morocco	7,899	2,514	1,020	633	697	1,821	2,063	53	−59	−72	−9.8	−60	−40	−8.0	−7.9	−22	−46
Mozambique	132	306	59	127	115	133	193	19	−98	−132	−67	−64	−6.9	−4.1	−28	−9.1	−9.6
Myanmar	1,300	707	779	(10)	(234)	315	(288)	(32)	−151	−43	−12	−53	−9.9	(51)	(−20)	(−7.0)	−19
Namibia	54	45	35	43	30	21	49	9.1	−14	−17	−12	−8.4	−1.2	−0.6	−4.1	−0.8	−1.3
Nauru	0.0	(0.0)	(0.0)	(0.0)	(0.0)	(0.0)	(0.0)	(0.0)	0.0	0.0	0.0	(0.0)	(0.0)	0.0	(0.0)	(0.0)	(0.0)
Nepal	1,562	701	2,126	310	427	381	424	93	−55	−244	−25	−189	−11	−31	−33	−10	−3.9
Netherlands	900	1,043	2,274	2,001	316	1,083	255	142	−40	−80	−4.7	−16	−6.7	−0.2	−2.1	−5.1	−1.8
New Zealand	542	294	551	336	85	226	49	24	−13	−39	−2.0	−14	−3.1	−1.9	−0.6	−8.1	−0.7

Country	IHD	Str	COPD	LRI	Diab	CKD	HHD	Card	Road	Self	Viol	Drwn	Mech	Natr	Anim	OTr	OUn
Nicaragua	41	(2.6)	(8.0)	(−0.7)	(−2.7)	36	(5.7)	(0.8)	−5.6	−4.1	−3.8	−1.9	−0.3	(6.4)	−0.3	(−0.3)	−0.3
Niger	202	(102)	(20)	(4.3)	(33)	(62)	(25)	(9.4)	−74	−17	−34	(−6.4)	(−6.8)	(0.0)	(0.5)	(−1.6)	(0.1)
Nigeria	1,364	(387)	(110)	(111)	(127)	(569)	(266)	(97)	−418	−178	−305	(−54)	(−27)	(−0.3)	(−12)	(−7.4)	(−5.6)
Niue	0.0	(0.0)	(0.0)	(0.0)	(0.0)	(0.0)	(0.0)	0.0	0.0	0.0	0.0	0.0	(0.0)	(0.0)	(0.0)	(0.0)	0.0
North Macedonia	373	594	132	48	97	64	219	71	−4.6	−8.8	−1.5	−2.8	−1.1	−0.1	−0.4	−1.7	−6.1
Northern Mariana Islands	0.2	(0.0)	(0.0)	(0.0)	(0.0)	(0.1)	(0.0)	(0.0)	0.0	0.0	0.0	(0.0)	(0.0)	(0.0)	(0.0)	(0.0)	(0.0)
Norway	267	291	608	669	54	174	49	18	−16	−51	−2.0	−8.9	−3.1	−0.6	−0.8	−8.0	−0.4
Oman	266	(30)	(15)	(−13)	(36)	(89)	(63)	(5.6)	−78	−1.4	−0.5	(−1.0)	(−2.3)	(0.0)	(0.0)	(−0.4)	(0.0)
Pakistan	14,556	4,231	4,252	968	2,859	3,803	1,951	584	−1,352	−672	−332	−704	−58	(−129)	−409	−96	−46
Palau	0.0	(0.0)	(0.0)	(0.0)	(0.0)	(0.0)	(0.0)	(0.0)	0.0	0.0	0.0	(0.0)	(0.0)	0.0	(0.0)	(0.0)	(0.0)
Palestine	421	139	52	43	131	154	121	8.6	−4.3	−3.1	−8.9	−6.0	−2.9	0.0	−0.3	−0.5	−1.4
Panama	11	4.0	6.2	(0.9)	(1.9)	8.6	3.4	(0.7)	−1.8	−1.5	−2.3	−1.3	−0.1	−0.1	−0.2	−0.1	−0.1
Papua New Guinea	429	357	491	139	197	64	70	(5.7)	−18	−11	−24	−101	−3.8	−16	−2.8	−6.2	−25
Paraguay	287	122	71	70	176	141	99	19	−37	−24	−28	−14	(−1.6)	−0.5	−2.2	(−1.0)	−3.4
Peru	985	612	515	3,145	368	1,511	232	13	−124	−49	−34	−82	−21	−15	−7.8	−38	−48
Philippines	612	(132)	(96)	(125)	(−8.9)	(259)	(212)	(27)	−108	−34	−108	−30	−3.4	(−8.2)	(−3.4)	(−4.8)	(−1.9)
Poland	5,264	3,319	1,561	4,422	936	886	1,984	1,176	−172	−230	−22	−75	−24	−0.7	−5.2	−48	−14
Portugal	621	890	1,111	1,361	325	702	346	34	−11	−71	−3.1	−16	−3.1	−0.6	−3.6	−15	−5.1
Puerto Rico	22	(1.1)	(7.8)	(3.7)	(−2.1)	(16)	(6.7)	(1.4)	−2.8	−1.9	−2.8	−0.2	(−0.1)	(1.5)	(−0.1)	(−0.1)	(0.0)
Qatar	153	(20)	(12)	(−7.7)	(53)	(76)	(17)	(2.2)	−48	−4.1	−0.7	(−0.4)	(−2.6)	0.0	(0.0)	(−0.8)	(0.0)
Republic of Korea	2,775	4,687	4,121	8,142	1,390	3,127	1,833	208	−194	−819	−18	−353	−28	−13	−13	−102	−18
Republic of Moldova	1,056	378	146	135	39	23	330	28	−9.2	−18	−5.3	−14	−2.2	−0.2	−1.2	−2.9	−4.5
Romania	3,547	3,292	1,034	1,379	197	413	2,318	963	−69	−81	−10	−74	−17	−1.0	−7.4	−30	−7.6
Russian Federation	30,980	25,650	6,052	4,722	2,386	2,107	2,355	2,789	−1,439	−2,601	−1,271	−586	−311	−11	−42	−280	−319
Rwanda	235	355	214	285	135	240	243	(10)	−23	−80	−16	−81	−11	−2.1	−49	−11	−23
Saint Kitts and Nevis	0.2	(0.0)	(0.0)	(0.1)	(−0.1)	(0.2)	(0.1)	(0.1)	0.0	0.0	0.0	(0.0)	(0.0)	(0.0)	(0.0)	(0.0)	(0.0)
Saint Lucia	0.2	(0.1)	(0.1)	(0.0)	(−0.1)	(0.3)	(0.2)	(0.1)	−0.1	0.0	−0.1	(0.0)	(0.0)	(0.0)	(0.0)	(0.0)	(0.0)
Saint Vincent and the Grenadines	0.2	(0.0)	(0.0)	(0.1)	(−0.1)	(0.1)	(0.2)	(0.0)	0.0	0.0	−0.1	(0.0)	(0.0)	(0.0)	(0.0)	(0.0)	(0.0)
Samoa	0.7	(0.1)	(0.1)	(0.1)	(−0.1)	(0.1)	(0.1)	0.0	−0.2	−0.1	0.0	−0.1	(0.0)	(0.0)	(0.0)	(0.0)	0.0
San Marino	2.4	2.3	1.8	2.7	0.5	1.2	1.6	0.4	(0.0)	−0.2	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Sao Tome and Principe	0.8	(0.1)	(0.3)	(0.5)	(0.0)	(0.6)	(0.2)	(0.2)	−0.3	0.0	−0.1	0.0	(0.0)	0.0	(0.0)	(0.0)	(0.0)
Saudi Arabia	2,991	514	(318)	(29)	(330)	1,926	(909)	(70)	−854	−110	−113	−31	−34	(−0.4)	(−3.6)	(−10)	(−16)
Senegal	90	38	(13)	(6.0)	(15)	38	(23)	(9.7)	−30	−19	−1.9	−6.6	−2.0	(−0.1)	(−2.4)	(−0.5)	(−1.3)
Serbia	1,675	1,648	545	260	264	356	680	255	−19	−63	−5.9	−13	−4.4	−0.3	−1.9	−11	−4.3
Seychelles	(0.3)	(0.0)	(0.0)	(0.4)	(−0.1)	(0.2)	(0.5)	(0.1)	−0.1	−0.1	0.0	(−0.1)	(0.0)	(0.0)	(0.0)	(0.0)	(0.0)
Sierra Leone	39	(11)	(2.6)	(12)	(−0.2)	(11)	(7.3)	(4.6)	−20	−5.8	−1.3	(−2.1)	(−1.2)	(0.2)	(−0.4)	(−0.9)	(−0.2)
Singapore	(−13)	(−7.5)	(−5.5)	(107)	(−2.9)	(−3.8)	(−1.2)	(3.5)	−2.8	−5.6	−0.5	(−0.2)	(−0.2)	0.0	(0.0)	(0.0)	(0.0)
Slovakia	1,340	541	202	561	78	157	230	53	−21	−32	−4.6	−15	−5.1	−0.4	−1.1	−9.2	−1.4
Slovenia	169	141	142	169	36	64	207	84	−4.9	−16	−0.5	−1.9	−1.8	−0.2	−0.6	−2.1	−0.1
Solomon Islands	1.4	(0.2)	(0.2)	(0.3)	(−0.1)	(0.1)	(0.1)	(0.0)	−0.3	−0.3	−0.1	−0.3	0.0	(0.0)	(0.0)	(0.0)	(0.0)
Somalia	34	25	(11)	(31)	(12)	22	13	(2.7)	−28	−9.2	−22	−9.7	−3.7	(1.0)	(−5.2)	(−1.2)	(−3.5)
South Africa	2,204	2,276	2,025	3,028	2,317	1,662	2,739	323	−164	−469	−554	−265	−53	−4.1	−111	−40	−41
South Sudan	20	(9.7)	(2.9)	(4.9)	(4.0)	(19)	(5.0)	(1.1)	−11	−4.8	−14	(−1.8)	(−1.1)	(0.0)	(−0.2)	(−0.3)	(−0.2)
Spain	3,190	2,501	5,998	2,876	870	2,903	1,659	419	−35	−205	−7.8	−86	−12	−2.8	−6.8	−21	−13
Sri Lanka	193	101	161	(−6.0)	(16)	(54)	51	(5.3)	−20	−38	−3.8	−13	−1.4	(0.7)	−4.3	(−1.8)	−1.3
Sudan	1,196	(161)	(55)	(−15)	(32)	(189)	(176)	(7.2)	−175	−31	−23	(−7.8)	(−19)	(−0.2)	(−0.6)	(−1.7)	(−2.3)
Suriname	0.8	(0.2)	(0.1)	(0.1)	(−0.1)	(0.5)	(0.2)	(0.1)	−0.3	−0.2	−0.1	(0.0)	(0.0)	0.0	(0.0)	(0.0)	(0.0)
Sweden	748	606	703	723	182	554	345	54	−26	−95	−8.8	−13	−7.6	−0.3	−1.1	−7.4	−2.7
Switzerland	682	402	605	682	133	667	535	57	−19	−56	−2.0	−8.6	−5.8	−1.7	−1.1	−9.9	−7.7
Syrian Arab Republic	4,210	752	408	226	250	829	684	25	−17	−8.2	−23	−18	−5.4	−0.4	−2.5	−9.5	−7.3
Taiwan	670	393	725	767	456	934	520	35	−67	−180	−4.1	−66	−3.5	−6.5	−0.9	−3.3	−10
Tajikistan	2,087	1,102	404	463	135	35	412	9.9	−36	−32	−9.9	−108	−2.3	−24	−0.1	−9.7	−3.1
Thailand	543	(267)	(201)	(−24)	(86)	(532)	(41)	(7.4)	−194	−55	−23	(−20)	(−3.3)	(−1.2)	(−1.0)	(−3.7)	(−0.8)
Timor-Leste	15	6.9	4.9	(0.4)	(−0.2)	(3.0)	4.2	(0.5)	−2.0	−1.6	−0.7	−2.6	(−0.1)	0.0	−0.3	(−0.1)	−0.4
Togo	32	(13)	(2.8)	(6.9)	(1.2)	(14)	(9.9)	(3.4)	−17	−6.9	−6.4	(−1.0)	(−0.9)	(0.0)	(−0.1)	(−0.3)	(−0.1)
Tokelau	0.0	(0.0)	(0.0)	(0.0)	(0.0)	(0.0)	(0.0)	(0.0)	0.0	0.0	0.0	(0.0)	(0.0)	0.0	(0.0)	(0.0)	(0.0)
Tonga	0.7	(0.1)	(0.2)	(0.1)	(0.1)	(0.1)	(0.0)	0.0	−0.1	−0.1	0.0	−0.1	(0.0)	(0.0)	(0.0)	(0.0)	0.0
Trinidad and Tobago	2.3	(0.2)	(0.1)	(0.2)	(−0.3)	(1.1)	(0.5)	(0.1)	−0.4	−0.3	−0.4	(0.0)	(0.0)	(0.0)	(0.0)	(0.0)	(0.0)
Tunisia	1,916	599	343	192	220	578	799	26	−21	−22	−2.2	−16	−18	−0.4	−2.6	−2.7	−7.0

Country	IHD	Str	COPD	LRI	Diab	CKD	HHD	Card	Road	Self	Viol	Drwn	Mech	Natr	Anim	OTr	OUn
Türkiye	9,064	5,727	9,370	4,863	2,950	4,795	5,571	135	−226	−178	−57	−97	−73	−82	−10	−72	−40
Turkmenistan	1,414	498	81	110	95	117	151	74	−4.1	−42	−8.5	−31	−4.5	−0.1	−1.4	−4.9	−7.9
Tuvalu	0.0	(0.0)	(0.0)	(0.0)	(0.0)	(0.0)	(0.0)	(0.0)	0.0	0.0	0.0	(0.0)	(0.0)	0.0	(0.0)	(0.0)	(0.0)
Uganda	271	208	141	177	115	203	192	(19)	−71	−132	−58	−107	−13	−1.8	−65	−13	−23
Ukraine	21,106	5,805	1,318	865	141	174	864	1,256	−201	−404	−130	−223	−37	−0.9	−6.3	−49	−38
United Arab Emirates	816	(126)	(93)	(−70)	(87)	(283)	(166)	(21)	−172	−12	−2.8	(−1.3)	(−7.8)	0.0	(0.0)	(−2.4)	(0.0)
United Kingdom	4,131	3,248	7,665	12,833	721	1,902	851	447	−89	−256	−12	−36	−14	−1.7	−5.0	−16	−5.4
United Republic of Tanzania	706	392	182	317	167	385	420	(33)	−92	−165	−88	−124	−10	−2.1	−61	−10	−16
United States of America	36,455	14,790	35,886	12,846	5,460	24,852	12,086	2,859	−1,239	−2,845	−628	−634	−194	−53	−64	−288	−51
United States Virgin Islands	0.9	(0.0)	(0.0)	(0.1)	(−0.1)	(0.2)	(0.2)	(0.1)	−0.1	0.0	−0.1	(0.0)	(0.0)	(0.0)	(0.0)	(0.0)	(0.0)
Uruguay	206	163	284	186	67	130	98	(13)	(−3.4)	−38	−5.9	−11	−3.7	−0.1	−0.6	−2.3	−3.8
Uzbekistan	11,906	3,001	601	918	943	862	1,028	73	−83	−231	−27	−253	−14	−0.9	−6.4	−16	−44
Vanuatu	5.6	(0.9)	0.9	(0.2)	(0.1)	(0.6)	0.4	0.1	−0.7	−0.9	0.0	−0.7	(0.0)	(−0.1)	0.0	(0.0)	−0.1
Venezuela (Bolivarian Republic of)	550	(60)	69	(8.4)	(16)	138	82	(5.7)	−67	−24	−63	−13	−7.6	(9.3)	−3.1	−3.9	−1.7
Viet Nam	2,119	3,190	1,208	(235)	807	1,321	864	155	−528	−304	−40	−318	−46	(−13)	−54	(−23)	−48
Yemen	2,162	674	256	(111)	77	206	608	(12)	−197	−67	−25	−57	−34	−3.7	−6.4	−13	−45
Zambia	268	250	97	130	94	229	255	35	−53	−116	−75	−72	(−5.3)	−0.2	−37	−11	−9.6
Zimbabwe	420	318	123	512	186	194	224	47	−31	−148	−44	−92	−8.5	−4.1	−46	−7.9	−23